LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy

A Capstone Project presented to the
College of Computer Studies of
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

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Bachelor of Science in Information Technology degree
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TABLE OF CONTENTS

TITLE

ADIVERS RECOMMENDATION SHEET

ACCEPTANCE SHEET

CERTIFICATION

ACKNOWLEDGEMENT

TABLE OF CONTENTS

LIST OF APPENDICES

LIST OF FIGURES

LIST OF TABLES

ABSTRACT

CHAPTER I: INTRODUCTION

- THE PROBLEM AND ITS BACKGROUND
- REVIEW OF RELATED LITERATURE
- SIGNIFICANCE OF THE STUDY
- STATEMENT OF THE PROBLEM
- SCOPE AND LIMITATIONS

CHAPTER II: METHODOLOGY

- INTRODUCTION
- RESEARCH DESIGN
  - ARCHITECTURAL FRAMEWORK
  - ALGORITHM

SOFTWARE ENGINEERING PARAGIDM

DATA FLOW DIAGRAM

TOOL TO TEST FOR ACCURACY

POPULATION AND SAMPLING

EVALUATION INSTRUMENT

DATA COLLECTION METHOD

ETHICAL CONSIDERATIONS

SUMMARY

CHAPTER III: RESULTS AND DISCUSSIONS

INTRODUCTION

PROJECT CAPABILITIES AND TESTING PROCESS DETAILS

PROJECT TESTING RESULTS

CHAPTER IV: CONCLUSIONS AND RECOMMENDATIONS

INTRODUCTION

CONCLUSION

RECOMMENDATIONS

REFERENCES

CURRICULUM VITAE

LIST OF APPENDICES

Appendix A : OPERATIONAL DEFINITION OF TERMS

Appendix B : EVALUATION SHEETS

Appendix C : PROJECT TIMELINE

Appendix D : WORK ASSIGNMENTS

Appendix E : CERTIFICATE OF APPRECIATION

Appendix F : SOURCE CODE

Appendix G : PLAGIARISM CHECK RESULTS

Appendix H : AI GENERATED CONTENT CHECK RESULTS

Appendix I : EXTENDED ABSTRACT

LIST OF FIGURES

Figure 1: Architectural Framework

Figure 2: Agile SDLC

Figure 3: Level 0 Context Diagram

Figure 4: Level 1 Main Process

LIST OF TABLES

Table 1: Population and Sample Distribution

Table 2: Title

Table 3: Title

ABSTRACT

Title : LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

The rapid growth of digital learning has made independent study an important part of modern education. Students increasingly use digital learning applications, such as review tools, flashcards, and quiz platforms, to support lessons outside the classroom. However, many existing educational platforms still rely on a one-size-fits-all learning approach, where the same learning activities and instructional materials are provided to all learners regardless of their individual learning progress or needs. Recent literature identifies personalized learning as a shift away from this traditional approach toward adaptive, learner-centered instruction (Fariani et al., 2023; Wang et al., 2025).

As a result, students may spend too much time reviewing concepts they already understand while receiving limited support for topics they find difficult. Learning activities that do not match students' knowledge and skill levels may also reduce engagement and contribute to boredom during independent study (Krannich et al., 2025). These challenges highlight the need for more personalized learning tools that can better address students' individual learning needs.

Cognitive Load Theory suggests that learning is more effective when instructional materials are designed to align with the learner’s limited working-memory capacity. In digital learning environments, excessive text, poorly

integrated multimedia, and distracting interface elements can impose irrelevant cognitive load, which reduces learning efficiency and weakens knowledge achievement (Surbakti et al., 2024; Krieglstein et al., 2023). Because learners differ in prior knowledge, attention, language proficiency, and learning ability, one-size-fits-all platforms are less effective than adaptive systems that create content, regulate pacing, and tailor presentation to individual needs (Surbakti et al., 2024).

Similarly, digital learning environments that adapt to students' learning needs provide more effective and engaging learning experiences by reducing unrelated cognitive load (Skulmowski & Xu, 2022). These findings highlight the importance of designing educational technologies that personalize learning according to students' knowledge and learning progress.

Digital learning technologies have improved opportunities for independent learning by giving students flexible access to educational resources anytime and anywhere. This flexibility allows learners to study at their own pace, adapt learning to their individual needs, and integrate education more easily into daily life (Sisouvong & Pasanchay, 2024; Josué et al., 2023). Digital platforms also provide learning tools such as quizzes, assignments, interactive resources, discussion spaces, and feedback systems that support active participation and continued engagement with course content (Alshamary & Alhalafawy, 2023; Josué et al., 2023).

However, many existing educational applications focus on only specific learning activities instead of providing a complete learning environment. Therefore, students may rely on several non-integrated applications to support different learning needs. Research has shown that this fragmented use of multiple learning platforms can increase unrelated cognitive load and tool fatigue, highlighting the importance of more integrated digital learning environments (Qiu & Qiu, 2026).

Integrating learning activities and learning analytics within a single digital platform provides a more effective learning experience and better supports students' self-regulated learning because it connects course participation, progress monitoring, reflection, and feedback in one environment. Studies show that platforms that embed self-regulated learning supports into digital learning systems improve students' achievement, digital competencies, self-regulation, and engagement compared with conventional platform use or traditional instruction (Elmabaredy & Gencel, 2024; Yang et al., 2024; Du et al., 2025). Learning analytics strengthens this effect by helping students observe their progress, identify weaknesses, plan next steps, and adjust learning strategies, especially when analytics tools are integrated with instruction and guidance rather than used as stand-alone tracking systems (Kleimola et al., 2024; Du et al., 2025).

Artificial Intelligence (AI) has become a promising technology for improving digital learning platforms by providing personalized and adaptive learning experiences. Unlike traditional educational applications that use fixed

learning activities, AI-powered systems can analyze students' learning progress, identify their strengths and weaknesses, and provide personalized learning paths, instructional content, and feedback based on their individual learning needs (Barrera Castro et al., 2024).

Recent research shows that AI-driven adaptive learning technologies improve learning experiences by adjusting instructional support based on students' performance and learning progress. This allows students to learn at their own pace while receiving personalized guidance (Bayly-Castaneda et al., 2024). These capabilities enable digital learning platforms to provide continuous and personalized support throughout the learning process, making learning more adaptive to each student's needs.

Pasig Christian Academy (PCA) is a private educational institution established in 1986 that offers Preschool, Elementary, and Junior High School education in Pasig City. Operating within the basic education sector, the institution is committed to fostering academic excellence and preparing its learners for higher education. This study specifically focuses on the Junior High School (JHS) department, where students gradually develop independent learning habits in preparation for the academic demands of Senior High School.

Based on the researchers' requirement analysis and consultation with the school administration, independent learning at Pasig Christian Academy primarily relies on official textbooks and teacher-provided instructional materials. These learning resources remain the primary references used by students during

independent study and effectively support the school's curriculum. However, these resources are primarily static, providing the same learning content and review materials to all students regardless of their individual mastery levels, learning pace, or specific learning needs. Furthermore, the researchers' needs assessment revealed that students experience difficulty identifying which topics require additional review, desire more engaging study activities, value continuous progress monitoring, and prefer a single platform that integrates multiple independent learning functions. These findings indicate opportunities to enhance the school's existing independent learning process through more personalized, adaptive, and engaging learning support.

Rather than replacing the school's existing instructional practices, these findings present an opportunity to enhance the current independent learning process through educational technology. LearnSync AI is proposed as an independent learning platform designed to complement the school's existing learning resources by providing personalized learning support, adaptive learning activities, AI-assisted study strategies, gamified learning activities, and learning analytics to support independent learning. The platform integrates Bayesian Knowledge Tracing (BKT) to estimate each student's knowledge mastery and Generative Artificial Intelligence (Generative AI) to generate personalized learning support based exclusively on educator-uploaded instructional materials. To maintain curriculum alignment and academic accuracy, only authenticated educators are authorized to upload learning resources and assign the

appropriate grade levels that may access them, while students receive AI-generated learning support derived solely from these verified materials.

LearnSync AI integrates several evidence-based study strategies within a single platform, including Adaptive Flashcards, Adaptive Test Mode, AI-assisted study strategies such as the Feynman Technique and Free Recall, an AI-powered learning assistant, AI-generated educational games, and learning analytics. Bayesian Knowledge Tracing (BKT) continuously estimates students' mastery levels based on their assessment performance, allowing Adaptive Flashcards and Adaptive Test Mode to adapt review activities according to each learner's estimated understanding. AI-generated educational games utilize these mastery estimates to tailor question difficulty, while the AI-assisted study strategies provide individualized feedback based on students' responses. Students also receive learning analytics through dashboards that display mastery progression, study streaks, digital badges, and performance statistics, encouraging greater self-awareness and independent learning.

Unlike conventional review materials that provide the same learning experience to every learner, LearnSync AI complements the school's existing independent learning process by integrating adaptive learning activities, AI-assisted personalized feedback, evidence-based study strategies, gamified learning activities, and learning analytics into a single platform. By combining educator-approved learning resources with Bayesian Knowledge Tracing (BKT) and Generative Artificial Intelligence (Generative AI), the proposed platform aims to provide a more personalized, engaging, and organized independent learning

experience while preserving the school's existing curriculum and instructional practices.

REVIEW OF RELATED LITERATURE

This chapter presents a comprehensive synthesis of local and international literature, research studies, and technological frameworks relevant to the development of LearnSync AI.

Modern Study Challenges and Diverse Student Learners

Developing an effective digital learning platform requires recognizing that students differ in how they learn, process information, and respond to instruction. Learners vary in their prior knowledge, academic abilities, motivation, learning pace, and study strategies, so a one-size-fits-all approach does not effectively support all learners (Castro et al., 2024; Gligorea et al., 2023). Research on personalized education suggests that learning environments become more effective when they consider individual learner characteristics and provide appropriate support (Castro et al., 2024). Likewise, adaptive learning has been identified as an alternative to traditional one-size-fits-all learning approaches because it provides learning experiences that respond to individual

learner needs and progress (du Plooy et al., 2024; Gligorea et al., 2023).

In addition, studies on adaptive learning emphasize that educational systems that respond to learners' needs and progress can create more engaging and effective learning experiences (du Plooy et al., 2024; Gligorea et al., 2023). As digital learning continues to expand, educational platforms are increasingly expected to provide flexible and personalized learning experiences that address the diverse needs of students (Castro et al., 2024).

Fast and Slow Learners

Students differ in the speed at which they process information, understand new concepts, and complete learning activities. These differences influence how learners respond to classroom instruction and digital learning environments. In mixed-ability classrooms, some students quickly master lessons and may become less engaged when instruction progresses too slowly, while others require additional time, practice, and guidance before fully understanding the material. Because learners have different academic needs, using the same teaching pace for everyone may reduce engagement and limit learning opportunities for

both high-performing and struggling students. Research on differentiated instruction suggests that learning environments should provide flexible support and varied learning opportunities to accommodate differences in students' abilities and learning pace (Ziernwald et al., 2022).

Likewise, research on educational differentiation indicates that adapting learning experiences to individual learner needs may improve participation, motivation, and academic achievement in diverse classrooms (Eikeland & Ohna, 2022). These findings highlight the importance of designing learning platforms that can support students at different levels of understanding rather than requiring all learners to progress at the same pace.

Digital Overload

The increasing use of digital technologies in education has created learning environments in which students often depend on multiple digital tools and platforms to access learning materials, complete assessments, submit assignments, and communicate with instructors and peers. Although these technologies can improve access and flexibility, managing different interfaces and constantly

navigating between separate systems can increase the mental effort required to complete learning tasks. This added effort can reduce the amount of cognitive capacity available for understanding, processing, and retaining new information, especially when students must divide their attention between learning content and platform navigation (Qiu & Qiu, 2026; Ibrahim et al., 2025).

Research on digital learning further suggests that unnecessary complexity in online environments contributes to extraneous cognitive load, meaning mental effort caused by the design of the learning system rather than by the learning material itself, which can reduce learning efficiency and make it more difficult for students to focus on instructional content (Nordin et al., 2026).

Furthermore, studies in learning analytics emphasize the importance of designing organized and accessible digital learning environments that support students through efficient access to educational resources and activities while minimizing unnecessary barriers to learning (Qiu & Qiu, 2026). Taken together, these findings highlight the value of developing integrated learning platforms that centralize academic resources and learning activities, thereby creating

a more organized, efficient, and student-centered learning experience.

Junior High School Students

Digital learning in the Philippines has increasingly required students to study with greater independence, especially in online and modular environments where teacher supervision is reduced and learners must manage their own time, tasks, and study strategies. Research shows that self-regulated learning is important in these settings because students differ in their ability to set goals, structure their learning environment, seek help, and monitor their progress, and weaker self-regulation is associated with poorer learning outcomes (Funa et al., 2023; Briones et al., 2023; Calamlam, 2023). At the same time, many Filipino learners continue to experience barriers such as limited internet connectivity, inadequate resources, difficulty understanding modules, and fragmented learning conditions, which can make independent learning more challenging (Bustillo & Aguilos, 2022; Agayon et al., 2022; Abejuela et al., 2023).

These conditions support the need for learning platforms that provide organized access to resources, adaptive support, and timely feedback. Studies on educational technology further suggest that students' learning experiences improve when digital systems are designed to be accessible, interactive, well-facilitated, and responsive to learner needs, including the use of adaptive learning systems and other technology-based interventions that support engagement and performance (Batucan et al., 2022; Llacuna et al., 2026; Caliston, 2025).

Thus, developing an integrated and adaptive learning platform for Junior High School students is well grounded in the literature, as this population is still strengthening independent learning habits and can benefit from personalized support, efficient resource access, and technology that aligns with their developmental and educational needs (Salva et al., 2022; Villarama et al., 2022; Castro et al., 2024).

The reviewed literature shows that students have different learning abilities, learning pace, and study habits. Because of these differences, a one-size-fits-all approach may not effectively support every learner.

Studies suggest that personalized and adaptive learning can better address individual learning needs by providing learning experiences that match each student's progress and level of understanding.

The literature also shows that students often rely on multiple digital tools and learning resources, which can make studying more difficult due to fragmented information and increased cognitive load. These findings highlight the need for a centralized learning platform that organizes learning resources and provides a more efficient and focused learning experience.

In addition, Junior High School students are still developing independent learning and self-regulated learning skills. Many learners experience challenges such as identifying their knowledge gaps, monitoring their learning progress, and maintaining effective study habits. Therefore, they can benefit from learning technologies that provide personalized guidance, adaptive learning activities, and timely feedback.

Overall, the reviewed literature supports the development of LearnSync AI as an independent learning platform for Junior High School students. By integrating Generative Artificial Intelligence (Generative AI) and Bayesian Knowledge Tracing (BKT), the platform aims to provide personalized learning support, centralized learning resources, and learning analytics to help students develop more effective independent learning habits.

Analysis of Independent Study Tools and Research Gaps

Independent study tools have become an important part of modern education because they give students flexible ways to review lessons and learn outside the classroom. However, many of these tools support only specific learning activities, which can lead students to rely on multiple applications and create a fragmented digital learning experience. Evidence shows that multi-platform learning environments increase extraneous cognitive load and tool fatigue, reducing students' ability to focus on actual learning tasks rather than on managing different technologies (Qiu & Qiu, 2026; Ibrahim et al., 2025).

Research also shows that structured instructional design and interactive learning environments can reduce cognitive load, suggesting that more integrated digital platforms would better support independent learning than disconnected single-purpose tools (Nordin et al., 2026). These findings reveal an important research gap in existing independent study tools, particularly the limited evidence on platforms that combine multiple evidence-based learning functions with adaptive support in a single, cognitively manageable environment.

Fragmentation of Current Educational Applications

Educational applications have become widely used to support independent learning by offering tools such as flashcards, quizzes, note-taking, and progress tracking. However, many of these applications are designed to support only specific learning tasks, so students often need to move across several platforms to complete different parts of their study routine. This creates a fragmented digital learning experience in which attention is divided between learning content and managing multiple technologies. Recent research shows that multi-platform learning environments increase extraneous cognitive load, contribute to tool fatigue, and make it harder for students to focus on meaningful learning tasks rather than navigation, logins, and platform switching (Qiu & Qiu, 2026; Ibrahim et al., 2025).

Research also suggests that fragmented learning resources often lack coherent organization, which further reduces learning efficiency and makes knowledge construction more difficult (Hu & Liu, 2025). Supporting this, integrated resource designs have been associated with better perceived coherence and lower cognitive load, indicating that more unified educational systems may better

support independent learning than disconnected single-purpose applications (Lou, 2025).

Limitations of Static Review Systems

Many educational applications still rely on static review systems that give all learners the same materials and review activities, even though learners differ in pace, prior knowledge, and current understanding (Bakri, 2025). This one-size-fits-all structure can be inefficient because conventional review often distributes the same materials uniformly rather than targeting each learner's actual weaknesses (Okubo et al., 2023) Research on personalized learning instead describes adaptive systems as ones that adjust to changing learner needs and recommend resources based on learner characteristics, knowledge level, and preferences (du Plooy et al., 2024; Abu-Rasheed et al., 2023).

Evidence also indicates that adaptive review can improve learning: a personalized review dashboard that recommended insufficiently understood pages produced a significantly higher improvement in quiz scores, while a broader review found that adaptive systems often improve

academic performance and engagement compared with non-adaptive environments (Okubo et al., 2023; du Plooy et al., 2024; Bakri, 2025).

The reviewed literature shows that many existing educational applications either support only specific learning activities or continue to rely on one-size-fits-all review approaches. As a result, students may experience fragmented learning environments and limited opportunities for personalized learning support. This fragmented approach can increase cognitive load, reduce learning efficiency, and make it more difficult for students to focus on understanding lessons. In addition, many current review systems continue to use a one-size-fits-all approach, providing the same learning materials and activities to all students despite differences in learning pace, prior knowledge, and individual learning needs.

Although existing applications offer useful learning features, the literature indicates a need for a more integrated and adaptive learning platform that combines personalized learning, organized learning resources, and continuous learning support within a single environment. Guided by these findings, LearnSync AI is proposed to address these research gaps by integrating Generative Artificial Intelligence (Generative AI), Bayesian Knowledge Tracing (BKT), adaptive learning activities, AI-assisted learning support, gamification, and learning analytics into one

centralized platform. Through these features, the platform aims to provide Junior High School students with a more personalized, organized, and engaging independent learning experience.

Learning Analytics and Student Motivation

Learning analytics has become an important educational tool because learner-facing dashboards give students direct access to information about their progress, helping them monitor performance and identify what they still need to study. Unlike static review systems that provide the same materials to all learners, dashboards can support self-regulated learning by making progress visible and guiding reflection, feedback, and next actions (Paulsen & Lindsay, 2024; Yang et al., 2024). Experimental and quasi-experimental evidence suggests that these systems can improve students' awareness of their learning, self-efficacy, engagement, and academic outcomes when the dashboards are intentionally designed to support motivation and regulation rather than simply display data (Fleur et al., 2023; Yang et al., 2024).

Research also indicates that dashboards work best when grounded in motivational theory and designed to help learners make autonomous, strategic decisions, highlighting the limitations

of static one-size-fits-all review approaches that do not adapt to students' ongoing progress (Lee & Kim, 2024;

Mastery Progress

Visualizing mastery progress helps students engage in self-monitoring because it makes their current standing and learning trajectory visible during the learning process, not only after formal grading events (Paulsen & Lindsay, 2024; Ponomariovienė et al., 2025). In competency-based settings, progress monitoring is tied to feedback, reflection, and self-regulation, with regular feedback helping students understand achievement and adjust strategies to reach learning goals (Ponomariovienė et al., 2025).

Learner-facing dashboards have increasingly shifted from teacher-oriented analytics tools to student-focused systems intended to support learning directly (Paulsen & Lindsay, 2024). A systematic review found that monitoring, awareness, and comparison are the most common dashboard affordances, but more recent designs increasingly include feedback, reflection, and goal-setting features that better support learning action (Paulsen & Lindsay, 2024).

Evidence from higher education also suggests that dashboards can increase engagement with self-assessment systems. In a study of 194 students, those with dashboard access interacted more with a computerized classification testing system and reported higher confidence than students using the same system without a dashboard (Ifenthaler & Şahin, 2023). The dashboard in that study displayed mastery-related indicators such as the number of mastered subject areas, correct and incorrect responses, and total attempts, showing how progress visualizations can make formative assessment more usable for students across a semester (Ifenthaler & Şahin, 2023).

Study Consistency and Behavioral Engagement

Study streaks and digital badges are features that can encourage students to build consistent learning habits by recognizing regular participation and achievement. A study streak tracks consecutive days of activity, while digital badges mark milestones such as completing lessons or meeting learning goals. These kinds of gamified feedback and recognition features tend to strengthen motivation, participation, and sustained involvement in digital learning

environments (Jun & Lucas, 2024; Sappaile et al., 2025; Fauzi et al., 2025).

Recent evidence also suggests that self-monitoring and feedback-linked gamification can support self-regulated learning and persistence. In one study, learners in a badge-sharing condition showed higher self-regulation and a higher completion rate than controls, with persistence of 70% versus 56% (Xu & Richardson, 2024). Broader reviews further indicate that badges, points, and feedback can promote active participation, progress tracking, and long-term involvement when they are meaningfully aligned with learning goals (Jun & Lucas, 2024).

Overall, current research supports the view that study streaks and digital badges can help promote regular participation, stronger engagement, and more consistent independent learning habits, although their effects depend on careful design and can be mixed for some learners (Thakarkar et al., 2025; Balci et al., 2022; Adanyin, 2024).

Time-on-Task and Self-Regulated Learning

Time-on-task information in digital learning environments can support students' self-regulated learning

by making study behavior visible and easier to evaluate during learning rather than only afterward. Clickstream evidence shows that online learners differ in time investment and time-use patterns across performance groups, suggesting that study time is a meaningful indicator of how students manage learning activities (Cao et al., 2022). More broadly, research on self-regulated learning tools shows that monitoring, goal setting, and self-evaluation are among the most commonly supported processes in online environments, which fits the use of time-on-task displays as a way to help learners track and adjust their behavior (Perez-Alvarez et al., 2022).

Recent dashboard studies further suggest that students value analytics displays for observing study progress and appraising their learning behavior, although their impact depends on clear reference points and integration into everyday learning processes (de Vreugd et al., 2024; Kleimola et al., 2024). Overall, displaying time-on-task appears to promote learner awareness, responsibility, and more intentional study habits, but the broader intervention literature shows that effects are mixed when analytics tools are not linked to planning and reflection (Heikkinen et al., 2022).

The reviewed literature demonstrates that providing students with direct access to their own learning data supports motivation and self-regulated learning by helping learners monitor their progress and study behaviors. Features such as visualizing mastery progress, tracking study time, and using gamified rewards like streaks and badges help learners build consistent study habits and monitor their own performance. These tools are especially important for students who need to manage their time and identify their own weaknesses during unsupervised study.

Guided by these findings, LearnSync AI features a personalized learning analytics dashboard designed to support Junior High School Students during their independent studies. The platform centralizes their educational data by displaying real-time mastery progress, study streaks, and digital badges. By transforming students' learning activities into clear, visual feedback, the platform encourages them to take greater responsibility for their own learning, maintain consistent study routines, and actively monitor their academic progress without requiring continuous teacher supervision.

Using Artificial Intelligence for Personalized Student Support

Artificial intelligence improves personalized student support by adapting instructional content, pacing, and feedback to individual learner needs rather than delivering the same materials to all

students. AI-based educational systems use learner data to build profiles, identify knowledge gaps, and recommend suitable learning activities, which helps students progress at their own pace and receive more targeted support (Barrera Castro et al., 2024; Murtaza et al., 2022).

Recent studies also suggest that adaptive feedback and intelligent tutoring systems can improve engagement, conceptual mastery, and satisfaction compared with static or traditional approaches, although effectiveness depends on design quality and subject context (Villegas-Ch. et al., 2025; Naseer & Khawaja, 2025; Sajja et al., 2023). At the same time, the literature consistently notes concerns about privacy, algorithmic bias, infrastructure, and the need to integrate AI with human teaching rather than replace it (Bahroun et al., 2023; Lin et al., 2023; Yarlagadda, 2025).

AI in Modern Classrooms

AI in modern classrooms has helped shift instruction from a uniform model toward adaptive learning environments that respond to individual learner needs. Recent reviews describe this transition as a departure from traditional, one-size-fits-all teaching, with AI enabling systems to analyze student performance, preferences, and interaction

data in order to tailor content, pacing, and instructional support (Lin et al., 2023; Rane, 2024; Yarlagadda, 2025). Adaptive platforms and intelligent tutoring systems use machine learning, natural language processing, and learning analytics to provide real-time feedback, targeted recommendations, and personalized learning pathways, which can make instruction more flexible and student-centered (Murtaza et al., 2022; Gligorea et al., 2023; Akintola et al., 2025).

Empirical studies also report gains in engagement and academic outcomes when AI-driven systems are implemented well, including a 25% improvement in grades, test scores, and engagement in one higher education experiment and higher post-assessment scores across mixed educational settings (Naseer et al., 2024; Sari et al., 2024; Villegas-Ch. et al., 2025). At the same time, the literature is clear that successful classroom integration depends on educator training, infrastructure, ethical safeguards, and maintaining AI as a support for human teaching rather than a replacement for it (Walter, 2024; Halkiopoulos & Gkintoni, 2024; Barrera Castro et al., 2024).

AI in Finding Student Learning Gaps

Artificial Intelligence has become an important tool for identifying students' learning gaps and providing personalized academic support. Unlike traditional assessments that often detect difficulties only after periodic testing, AI-driven adaptive learning systems can continuously analyze student performance and learning behaviors to identify weaknesses in real time and provide immediate support (Akintola et al., 2025; Onesi-Ozigagun et al., 2024; Khanipoor & Karimian, 2024).

These systems can then recommend targeted learning activities, adjust content difficulty, and deliver personalized feedback, helping students focus on areas that need improvement and strengthen understanding during independent learning (Akavova et al., 2023; Gupta, 2024).

The reviewed literature highlights the important role of Artificial Intelligence in enhancing education by providing personalized learning experiences, adapting instructional content based on students' performance, and identifying learning gaps that require additional support. These capabilities address the limitations of traditional learning

approaches, which often provide the same learning experience to all students regardless of their individual needs and progress.

Guided by these findings, LearnSync AI integrates Artificial Intelligence to deliver adaptive and personalized academic support for Junior High School Students. By continuously adjusting learning activities according to students' progress and identifying areas that require reinforcement, the platform promotes more effective independent learning and encourages deeper conceptual understanding. Through this adaptive approach, LearnSync AI aims to provide a responsive learning environment that improves student engagement, strengthens knowledge retention, and supports better learning progress.

Combining Gamification and Evidence-Based Learning Strategies

Gamification and evidence-based learning strategies appear to support different but complementary aspects of learning in digital platforms. Gamification primarily improves motivation, engagement, and participation, while evidence-based strategies such as retrieval practice, repeated practice, and immediate feedback are more directly tied to stronger retention and conceptual learning (Nylén-Eriksen et al., 2025; Andretta et al., 2025; Trayco et al., 2025). Recent reviews and empirical studies suggest that the most effective gamified environments are those that deliberately align

game mechanics with sound instructional design, feedback, and opportunities for active recall rather than using rewards as isolated features (Dahalan et al., 2023; Lee et al., 2025; Zatserkovnyi, 2025).

Emerging work further suggests that AI-enhanced gamified systems may strengthen this integration by adapting difficulty, feedback, and reward timing to learners' progress while supporting long-term retention and motivation (Kumar, 2026; Aswini et al., 2026; Kassenkhan et al., 2025).

Gamification Procedures

Gamification has become an effective teaching approach that can improve student engagement, motivation, and learning outcomes by incorporating game elements into educational activities. Recent studies show that gamification can increase learners' participation, motivation, and performance in both classroom and digital learning environments (Dahalan et al., 2023; Khaldi et al., 2023; Dehghanzadeh et al., 2023; Hellín et al., 2023; Qudsi, 2024). In digital learning platforms, gamified activities provide students with interactive opportunities to review previously

learned concepts while sustaining engagement during independent study.

Research also suggests that gamification is most effective when game elements are aligned with sound instructional design, meaningful feedback, and clear educational objectives rather than functioning only as entertainment or reward systems (Dehghanzadeh et al., 2023; Khaldi et al., 2023; Trayco et al., 2025; Kode, 2025). However, the literature also notes that effects can vary depending on learner characteristics, novelty effects, and the quality of implementation (Ratinho & Martins, 2023; Ukgoda, 2025).

Shared Mechanism

Crossword puzzles, matching cards, and timed trivia questions are connected because they all use the same basic learning process: students must remember and retrieve information, not just reread it, and gamification research shows that this kind of activity can improve learning, motivation, and participation (Monzon & Hays, 2024; Dahalan et al., 2023). They also share common game features such as challenge, immediate feedback, and clear

goals, which help keep students engaged during review activities (Al-Khresheh, 2025). Crossword studies show that puzzles improve terminology recall, vocabulary retention, and active participation because students solve clues by remembering meanings and spellings (Jamal et al., 2026; Ramadhani & Shofiyuddin, 2025; Hsu et al., 2023). Matching card activities show a similar pattern because students must connect related ideas from memory, and card-based learning has been found to improve retention, interest, and confidence (Yu et al., 2025; Gursesli et al., 2023).

Timed trivia is related to the other two because it also depends on fast recall, repeated practice, and immediate correction of answers, which are features linked to better memory and attention in gamified learning (Al-Khresheh, 2025). In simple terms, the three games are different in format, but they do the same job: crossword strengthens word and term recall, matching cards strengthen associations between related concepts, and timed trivia strengthens quick retrieval and feedback-based review (Jamal et al., 2026; Yu et al., 2025; Hsu et al., 2023).

Crossword

Crossword puzzles have become a useful educational activity for reinforcing knowledge and improving students' understanding of key concepts. Unlike traditional review methods that rely only on reading or memorization, crossword puzzles encourage learners to actively recall information by identifying and connecting terms with their corresponding definitions or clues. This process promotes deeper cognitive engagement while making learning more interactive and enjoyable.

Research suggests that crossword puzzles can improve students' understanding of key concepts and support better performance in objective assessments by encouraging active recall during learning activities (Arnold et al., 2024). These findings suggest that incorporating crossword puzzles into digital learning platforms can strengthen concept retention while increasing student engagement during independent study.

Matching Card Games

Matching card games are widely used as interactive learning activities that promote active participation and reinforce students' understanding of concepts. By requiring learners to match related terms, definitions, images, or concepts, these activities encourage active recall and strengthen associations between pieces of information rather than relying on passive reading.

Recent research found that matching card-based learning materials significantly improved students' vocabulary mastery and learning outcomes by providing an engaging and interactive learning experience (Febrianti & Shofiyuddin, 2025). Because matching activities encourage learners to repeatedly recall and associate related concepts, their underlying principles can be adapted to reinforce foundational knowledge in digital learning environments.

Timed Trivia Questions

Timed trivia questions are an effective gamified learning activity that combines retrieval practice with

interactive quiz-based learning. By requiring learners to answer review questions within a limited time, these activities promote active recall while maintaining attention under structured constraints. Recent evidence shows that game-based quiz platforms improve student engagement, motivation, participation, and academic performance across different educational settings (Dahalan et al., 2023; Nadeem et al., 2023; Özdemir, 2024; Raje & Tamilselvi, 2024).

In addition, game elements such as time limits, scores, leaderboards, and immediate feedback appear to strengthen participation and learning when they are aligned with instructional goals (Jaramillo-Mediavilla et al., 2024; My et al., 2024; Ratinho & Martins, 2023; Ningsih, 2025). Together, these findings support timed trivia questions as an engaging review strategy that encourages active participation while reinforcing previously learned concepts through repeated retrieval and formative feedback (Pellas, 2024; Zainuddin, 2023).

Retrieval Practices

Cognitive retrieval practices are an important component of modern educational psychology because they enhance how students recall and retain learned information. Unlike traditional study methods that rely on rereading or passive memorization, retrieval-based learning requires students to actively recall information from memory, which strengthens long-term retention and improves understanding of concepts (Roelle et al., 2022; Thompson & Hughes, 2023; Serra et al., 2025; Madan, 2023).

Research across educational settings shows that retrieval practice is generally more effective than passive review strategies for improving learning and delayed retention, and it can also support transfer, problem solving, and metacognitive monitoring (Corral & Carpenter, 2025; Trumble et al., 2023; Muzsnay et al., 2025). Retrieval practice can be implemented through strategies such as free recall, low-stakes testing, and structured explanation exercises such as the Feynman Technique, which encourage learners to reconstruct knowledge actively from memory (Wellmann & Torres Skillicorn, 2023; Linh & Oanh, 2025; Rajesh & Khan, 2024).

Feynman Technique

The Feynman Technique is a learning strategy that helps students develop a deeper understanding of concepts by explaining material in their own words using simple and clear language. Explanation-based learning tasks prompt learners to generate coherent verbal statements that clarify, interpret, and justify ideas, which helps integrate new information with prior knowledge and supports knowledge generalization (Fiorella, 2023). Rather than relying on memorization alone, this kind of learning requires active reconstruction of knowledge, and retrieval-oriented explanation tasks can strengthen both comprehension and long-term retention (Roelle et al., 2023; Roelle et al., 2022).

The Feynman Technique also encourages learners to identify knowledge gaps and revisit source material until they can explain a concept accurately and simply (Linh & Oanh, 2025; Rajesh & Khan, 2024). This process aligns with research showing that generative activities mainly support the construction of coherent mental representations, while retrieval practice mainly supports their consolidation in

memory (Roelle et al., 2023; Roelle et al., 2022; Obergassel et al., 2025). Recent experimental evidence suggests that combining generative tasks and retrieval tasks produces better retention than generative tasks alone and better comprehension than retrieval tasks alone, which fits the logic of the Feynman Technique particularly well (Obergassel et al., 2025; Roelle et al., 2023).

Free Recall

Free recall is a form of retrieval practice that helps students strengthen their memory by encouraging them to remember what they have learned without using notes or other learning materials. Instead of simply rereading information, students actively retrieve knowledge from memory, which strengthens later recall and supports long-term retention (Rea et al., 2022; Serra et al., 2025). Research shows that retrieval practice, including free recall, is generally more effective than passive study methods such as rereading or repeated study for

improving delayed learning and memory (Aljabri, 2024; Bego et al., 2024; Klier & Buratto, 2023).

Repeated retrieval can also support recognition, conceptual retention, and transfer when practice is sufficiently repeated and assessed after a delay (Corral & Carpenter, 2025; Huang & Cao, 2026). These findings suggest that free recall is an effective cognitive learning strategy for supporting students' independent learning, although benefits depend on task demands and implementation (Redifer et al., 2025; Lima & Buratto, 2024).

Adaptive Flashcards

Adaptive flashcards are an effective learning tool because they combine active recall with personalized review scheduling. Unlike passive study methods such as rereading notes, flashcards require learners to retrieve information from memory before checking the answer, which strengthens recall and reinforces learned concepts (Huang et al., 2025; Jape et al., 2022). Research on vocabulary learning shows that flashcards can improve recall performance and learning outcomes, and digital flashcards can outperform both paper

flashcards and word lists in structured learning contexts (Le & Luong, 2023; Xodabande et al., 2022). Their effectiveness can be further enhanced through spaced repetition, which schedules review sessions over time and increases long-term retention and knowledge transfer compared with non-spaced review or lecture-based study (Price et al., 2024; Santhosh et al., 2024).

Modern adaptive systems extend this approach by adjusting review intervals according to learner performance and item difficulty, which helps focus attention on difficult concepts while reducing unnecessary repetition of well-learned material (Huang, 2025; Jape et al., 2022; Xiao & Wang, 2024).

Adaptive Assessment

Computerized adaptive testing has become an important development in modern educational assessment, shifting traditional fixed testing formats toward more flexible and data-driven evaluation systems. Unlike conventional assessments where all students answer the same set of questions regardless of ability, adaptive testing systems adjust question difficulty based on learners' responses and

performance patterns. Recent research highlights that these systems use real-time performance data and machine learning techniques to select appropriate test items, allowing assessments to better match each student's level of understanding (Zhuang et al., 2024).

In addition, AI-driven quiz generation systems have been shown to support the creation of dynamic assessments that can adjust question difficulty and provide more personalized evaluation experiences (Al-Gburi, 2025). Through these adaptive mechanisms, computerized testing systems aim to improve the efficiency of assessments and provide more accurate representations of student knowledge and learning progress.

The reviewed literature suggests that integrating gamification, retrieval-based learning strategies, adaptive flashcards, and adaptive assessment can contribute to more effective and engaging learning experiences compared to traditional passive study methods. Gamification has been shown to enhance student motivation and engagement by incorporating interactive and goal-oriented learning activities, while cognitive strategies such as active recall and retrieval practice support deeper understanding and long-term retention of information. Likewise,

spaced repetition and adaptive flashcards promote more efficient review by prioritizing concepts that require additional practice while reducing unnecessary repetition of concepts that learners have already mastered. In addition, adaptive assessment approaches improve learning efficiency by adjusting the difficulty and content of evaluation tasks based on student performance, allowing learners to be assessed according to their current level of understanding.

Collectively, these approaches highlight the value of combining motivational, cognitive, and adaptive learning strategies within digital learning environments. Guided by these findings, LearnSync AI integrates spaced repetition, Bayesian Knowledge Tracing–based adaptive flashcards, adaptive assessments, retrieval-based learning strategies, and gamified learning activities into a unified platform that supports personalized, adaptive, and self-directed learning. Through this integrated approach, the platform aims to promote student engagement, reinforce knowledge retention, and support independent learning.

Performance Evaluation of Language Models and Inference Systems

The performance of large language models and their inference systems is an important factor in ensuring that AI-powered educational platforms provide fast, reliable, and responsive learning experiences. In digital learning environments,

delays in processing user inputs or generating responses can negatively affect student engagement and disrupt learning continuity. As a result, modern LLM systems are designed with optimized inference and serving architectures that improve how models handle multiple user requests efficiently. Research on distributed model serving systems shows that optimized inference techniques can improve system throughput and response efficiency in transformer-based language models (Yu et al., 2022).

In addition, studies on memory-efficient LLM serving highlight that improved resource management and computational optimization help reduce latency and support smoother real-time interactions in large-scale AI applications (Kwon et al., 2023). These advancements are particularly important for educational platforms that rely on generative AI, as they ensure timely feedback, stable performance, and uninterrupted learning support for users.

LLM in Materials and Context

Large language models (LLMs) play an increasingly important role in helping students engage with complex and lengthy learning materials. Early transformer-based language models showed strong capabilities in

understanding and generating human-like text by learning semantic, syntactic, and contextual relationships across language inputs (Shi et al., 2024; Yan et al., 2023). As these models scaled, they developed stronger in-context learning, instruction-following, and reasoning abilities, which expanded their potential use in educational systems as reading, writing, and explanation assistants (Shi et al., 2024; Wang et al., 2024). Although earlier systems were constrained by relatively limited context windows, recent advances in long-context modeling have extended LLM capacity to process much longer inputs, improving their ability to maintain coherence across multi-topic materials, long documents, and extended interactions (Chen et al., 2023; Xu et al., 2023; Yen et al., 2024).

In educational settings, these improvements support more personalized explanations, adaptive pacing, real-time feedback, and better synthesis of dense materials during independent study (Sharma et al., 2025; Wang et al., 2024; Zhui et al., 2024). However, the evidence also shows that effective support depends not only on larger context windows, but on retrieval, context management, and careful instructional design, given persistent risks related to

hallucinations, bias, privacy, and student overreliance (Mei et al., 2025; Xu et al., 2023; Yan et al., 2023).

Real Time Feedback

Large language model–based systems play an important role in providing immediate formative feedback within digital learning environments. In traditional systems, delays in generating responses may interrupt students' focus during active learning tasks such as retrieval practice and review activities. In contrast, modern AI-powered systems are capable of delivering near real-time feedback and personalized guidance, which supports student engagement and sustained participation in learning activities (Kasneci et al., 2023).

Reducing response delays also helps create a smoother interaction between the learner and the system, minimizing disruption during study sessions. This immediate feedback allows students to quickly identify and correct mistakes, which can improve learning efficiency, task engagement, and revision behavior (Meyer et al., 2024).

Model Specialization and Dynamic Routing

The use of multiple large language models within a single system plays an important role in improving the efficiency and responsiveness of AI-powered applications. Instead of relying on a single model to handle all types of tasks, modern systems can distribute requests by assigning them to different models based on task complexity and computational requirements. Research on LLM routing systems shows that queries can be directed to different models depending on expected performance needs, allowing systems to balance quality and efficiency (Ong et al., 2024).

In this approach, simpler or less computationally demanding tasks can be handled by more efficient models, while more complex requests are processed by higher-capacity models. This routing strategy helps optimize system performance by reducing unnecessary computational load and improving response time, resulting in a smoother and more efficient user experience in real-time applications.

The reviewed studies demonstrate that an effective AI-powered educational platform depends not only on the capabilities of large language models but also on efficient inference systems, real-time

feedback, contextual understanding, and appropriate task distribution among AI components. Together, these technologies create a learning environment that is responsive, reliable, and capable of supporting students throughout their independent study sessions.

Guided by these findings, LearnSync AI integrates these concepts to provide fast AI-generated assistance, process lengthy study materials effectively, deliver immediate personalized feedback, and maintain smooth system performance. By combining this into a unified application, it provides Junior High School Students with a more efficient, engaging, and personalized learning experience that supports independent study and improves overall learning outcomes.

Bayesian Knowledge Tracing

Bayesian Knowledge Tracing (BKT) is a probabilistic model used in intelligent tutoring systems to estimate a learner's mastery of specific skills based on their sequence of correct and incorrect responses. It works by continuously updating the probability that a student has learned a concept, making it useful for modeling knowledge progression in adaptive learning environments.

BKT is commonly used in educational systems due to its simplicity, interpretability, and ability to support dynamic learner

modeling without requiring high computational complexity (Bulut et al., 2023).

Tracing Models: BKT, DKT, and PFA

Bayesian Knowledge Tracing (BKT), Performance Factors Analysis (PFA), and Deep Knowledge Tracing (DKT) remain central approaches in educational data mining for modeling student learning progression, but they differ substantially in representation, predictive behavior, and practical use. BKT models student mastery as binary latent knowledge states updated over time through a Hidden Markov framework, which makes it especially useful for tracking learning sequentially and for producing transparent estimates of mastery (Bulut et al., 2023; Bai et al., 2024). By contrast, PFA is a logistic alternative to traditional knowledge tracing that predicts future performance from prior successes and failures on one or more knowledge components, making it flexible for items tagged with multiple skills and competitive in some contemporary settings (Lee et al., 2025; Zhang et al., 2025). DKT, in contrast, applies recurrent neural networks to exercise-response sequences and is widely used because it captures complex student interaction

dynamics and often achieves strong predictive performance, with later variants such as AT-DKT reporting improved AUC and accuracy across real-world datasets (Lee et al., 2025; Liu et al., 2023).

However, the literature consistently identifies a trade-off between accuracy and interpretability: deep learning knowledge tracing models can model complex behaviors effectively, particularly when large training datasets are available, but they are often opaque and require substantial tuning, whereas BKT and other transparent models are easier to interpret in educational applications (Bulut et al., 2023; Bai et al., 2024; Zhou et al., 2024). At the same time, simpler models remain practically important because BKT continues to support personalized intervention through interpretable probabilistic parameters, and recent evidence suggests that PFA can remain competitive while BKT may be more stable than more complex models under concept drift (Bulut et al., 2023; Lee et al., 2025). Overall, recent research suggests that selecting among BKT, PFA, and DKT involves balancing predictive power, transparency, scalability, and robustness, rather than identifying a single best model for all educational contexts

(Bai et al., 2024; Zhou et al., 2024; Abdelrahman et al., 2022)

BKT Selection

Among the different knowledge tracing models reviewed, including Performance Factors Analysis (PFA) and Deep Knowledge Tracing (DKT), the researchers selected Bayesian Knowledge Tracing (BKT) as the learner modeling approach for the system. This decision is based on its suitability for adaptive learning environments that require efficient, interpretable, and sequential tracking of student knowledge.

Unlike DKT, which requires large datasets and higher computational resources, and PFA, which focuses on aggregated performance counts rather than step-by-step learning progression, BKT provides a lightweight framework that updates learner mastery based on individual responses. Its simplicity and transparency make it appropriate for educational applications where clear interpretation of student progress is important. Therefore, BKT was chosen to support the system's need for continuous, real-time, and

interpretable tracking of learner performance in adaptive learning activities such as flashcards and assessments.

The integration of Bayesian Knowledge Tracing (BKT) into LearnSync AI provides an efficient method for tracking student mastery during independent study sessions. By continuously updating a learner's knowledge state based on their responses, the system can estimate skill mastery while accounting for random factors such as guessing and slipping. Its lightweight learner modeling approach also helps maintain fast and responsive system performance.

Furthermore, BKT enables LearnSync AI to personalize adaptive learning activities by adjusting the difficulty of the Adaptive Flashcards and Adaptive Test Mode according to the learner's estimated mastery level. This supports a more individualized independent learning experience while maintaining an interpretable and computationally efficient learner model.

SIGNIFICANCE OF THE STUDY

This study proposes LearnSync AI, a personalized learning platform designed to enhance independent learning through personalized learning support, adaptive learning activities, AI-assisted study strategies, gamified learning experiences, and learning analytics. The proposed system is expected to benefit the following:

Pasig Christian Academy. The proposed system will be a major advantage for Pasig Christian Academy by providing a centralized, AI-assisted learning platform that complements the school's existing independent learning process. LearnSync AI enables the institution to integrate adaptive learning, AI-assisted learning support, personalized assessments, and verified instructional materials into a secure learning environment. Because the platform is designed strictly for asynchronous, independent study at home, it acts as a modern extension of the curriculum without disrupting the school's traditional, face-to-face daily operations.

Educators. The proposed system provides educators with a secure and centralized platform for uploading and managing official learning resources used by LearnSync AI. By ensuring that AI-generated learning support is based only on teacher-approved instructional materials, the system helps maintain curriculum alignment and academic accuracy. In addition, the educator dashboard allows teachers to view students' learning progress and mastery

statistics, enabling them to gain insights into students' independent learning without significantly increasing their workload.

Junior High School Students. The proposed system provides Junior High School students with a personalized study companion that customizes learning activities based on their current skill levels. By combining adaptive learning activities, AI-assisted feedback, gamified learning experiences, and personalized learning analytics in a single platform, the LearnSync AI encourages students to become more engaged in independent learning while helping them monitor their mastery progression and strengthen their study habits. This ensures students truly understand their key lessons and build strong study habits as they prepare for the academic demands of Senior High School.

Researchers. The study will contribute to the development of the researchers' knowledge and skills in the fields of software development, educational technology, artificial intelligence, adaptive learning systems and Bayesian Knowledge Tracing. It will also give them good practice in designing, developing, implementing and evaluating an AI-based educational platform.

Future Researchers. The findings of this study may serve as a reference for future researchers developing AI-powered educational platforms. It may also provide a foundation for further studies on adaptive learning, Generative Artificial Intelligence, Bayesian Knowledge Tracing, and personalized learning technologies across different educational contexts.

STATEMENT OF THE PROBLEM

As Junior High School students progress through increasingly complex academic subjects, independent learning becomes essential in reinforcing classroom instruction and preparing them for the academic demands of Senior High School. At Pasig Christian Academy, students primarily conduct independent study using educator-provided learning materials and official textbooks. While these resources effectively support the school's curriculum, the current independent learning process provides limited opportunities for personalized learning, adaptive learning support, and continuous monitoring of students' learning progress. To complement the school's existing independent learning process, the researchers propose LearnSync AI, an educational platform that integrates Generative Artificial Intelligence (Generative AI) and Bayesian Knowledge Tracing (BKT) into a single platform that provides personalized learning support, adaptive learning experiences, and continuous monitoring of students' learning progress.

This study seeks to evaluate the platform's technical accuracy, support for independent learning, changes in students' learning performance, and overall acceptability through the following research questions:

1. What are the current independent learning practices and challenges of the Junior High School students of Pasig Christian Academy?

2. What features and functionalities, including the integration of Generative Artificial Intelligence and Bayesian Knowledge Tracing, can be incorporated into the proposed system to address the identified challenges and requirements?

3. How can the correctness of the implemented Bayesian Knowledge Tracing engine be verified against manually computed mastery estimates?

4. How significant is the difference between the students' standard pre-test scores and their parallel post-test scores after using LearnSync AI?

5. What is the level of acceptability of LearnSync AI as evaluated by the intended users, specifically Junior High School students and educators based on the functionalities accessible to them, in terms of the following eight ISO/IEC 25010 software quality characteristics:

    - Functional Suitability
    - Performance Efficiency
    - Compatibility
    - Usability
    - Reliability
    - Security

- Maintainability
- Portability

SCOPE AND LIMITATIONS

This study focuses on the design, development, implementation, and evaluation of LearnSync AI, an independent learning platform that supports the independent learning of Junior High School students through adaptive learning, Generative Artificial Intelligence (Generative AI), Bayesian Knowledge Tracing (BKT), gamification, and learning analytics in a single platform.

Scope of the Study:

This study is limited to the implementation and evaluation of LearnSync AI involving 50 selected Junior High School students (Grades 7 - 10), representing 50% of the identified accessible Junior High School population (100 students), from which the respondents will be selected. of Pasig Christian Academy during the Academic Year 2026–2027. Although the respondents come from different grade levels, they will be analyzed as a single study population. Comparative analyses among grade levels are outside the scope of the study. Instead, the evaluation focuses on

determining the system's effectiveness by analyzing the significant difference between the students' standard pre-test scores and their parallel post-test scores.

The platform is designed to complement the school's existing learning process by supporting students' independent study outside regular classroom instruction. It is not intended to replace traditional classroom teaching or disrupt scheduled learning activities. LearnSync AI serves as an extension of the school's curriculum by providing AI-assisted learning experiences using educator-approved instructional materials.

The platform includes an educator module that enables authenticated educators to upload and manage official learning resources and view student learning statistics. These uploaded materials serve as the exclusive knowledge base for all AI-assisted functionalities, ensuring that generated content remains aligned with the school's curriculum while maintaining academic accuracy and consistency.

LearnSync AI integrates the classical four-parameter Bayesian Knowledge Tracing (BKT) model to estimate each learner's mastery level based on their performance in Adaptive Flashcards and Adaptive Test Mode. These mastery estimates are then used to personalize learning activities and dynamically adjust

the question selection and difficulty of AI-generated educational games, including Crossword, Term Matching, and Timed Trivia.

The platform further supports independent learning through Generative Artificial Intelligence (Generative AI) by providing AI-assisted study strategies such as the Feynman Technique and Free Recall, personalized feedback, and an AI-powered learning assistant capable of answering questions based exclusively on educator-uploaded learning resources.

Finally, the study evaluates LearnSync AI using the applicable ISO/IEC 25010 Software Quality Model, specifically the characteristics of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Limitations of the Study:

The implementation and evaluation of LearnSync AI are limited to a 2 to 4 week period involving 50 selected Junior High School students of Pasig Christian Academy. The testing and implementation phase of the system will not cover an entire academic year's curriculum due to the time constraints of the development period. Because of this limited implementation period, the study primarily evaluates the extent to which LearnSync AI

supports students' independent learning and examines short-term changes in learning performance through the students' pre-test and post-test results and BKT mastery estimates. The study does not measure long-term knowledge retention, sustained learning behaviors, long-term academic achievement, or permanent changes in student performance.

To ensure a manageable implementation, the learning activities will initially focus on the subject identified through the requirements analysis as the most appropriate for the current capabilities of LearnSync AI for each participating grade level. Although respondents come from different grade levels, the study does not perform comparative analyses among Grades 7, 8, 9, and 10. Instead, all participants are analyzed as a single study population.

The study employs a one group pre test/post test design without a separate control group. Therefore, while the comparison of pre-test and parallel post-test scores can determine whether a significant change in students' learning performance occurred following the implementation of LearnSync AI, the study cannot establish that LearnSync AI alone caused the observed change. Other factors, such as classroom instruction, independent study outside the platform, prior knowledge, and exposure to other

learning resources, may also contribute to changes in student performance.

The current implementation of LearnSync AI is primarily designed for concept-based learning and review. Specialized Mathematics instruction involving symbolic computation, complex equation solving, graphing, and detailed step-by-step mathematical problem-solving is outside the scope of the platform. When Mathematics is identified as a difficult subject for a participating grade level, the study may instead focus on another highly identified subject that is compatible with the current concept-review capabilities of LearnSync AI. Specialized Mathematics learning functionality may be considered for future development.

The platform employs only the classical Bayesian Knowledge Tracing (BKT) model. Student mastery estimates are updated exclusively through their performance in Adaptive Flashcards and Adaptive Test Mode. AI-generated educational games utilize existing mastery estimates only to personalize question selection, generated content, and difficulty level without modifying the learner's mastery state. Likewise, the AI-assisted Feynman Technique and Free Recall provide personalized explanations and feedback but do not directly update BKT mastery estimates.

Since the Bayesian Knowledge Tracing (BKT) model initially has no prior evidence of a learner's knowledge state, all students begin with the same default mastery probabilities across all learning concepts. Consequently, personalization is not immediate and becomes progressively more representative of each learner's actual knowledge as they complete Adaptive Flashcards and Adaptive Test Mode throughout the implementation period.

The AI-powered learning assistant generates responses exclusively from educator-uploaded learning resources. Consequently, the quality, completeness, and accuracy of AI-generated explanations depend entirely on the uploaded instructional materials. The chatbot cannot generate information beyond the available learning resources or external knowledge sources.

Operationally, LearnSync AI requires a stable internet connection for all online functionalities. The platform accepts plain-text PDF learning modules with a maximum file size of 20 MB. To accommodate the school's continued reliance on printed textbooks and other physical instructional materials, the platform also supports image-based or scanned documents with a maximum of 20 pages, which are processed using Optical Character Recognition (OCR) for use as AI learning resources. The accuracy of the OCR process depends on the quality of the uploaded images

or scanned documents. Blurred, low-resolution, poorly illuminated, skewed, or otherwise unclear documents may reduce text recognition accuracy, which may consequently affect the quality of AI-generated learning support.

To preserve academic integrity and ensure curriculum alignment, only authenticated educators are authorized to upload learning resources. Students are not permitted to upload instructional materials, and all AI-assisted functionalities depend entirely on the availability of educator-provided learning resources. The platform does not support offline functionality, handwritten text recognition, voice-based interaction, payment processing, automated billing, or enterprise-level server scalability.

The platform depends on third-party services for its core functions, including the Gemini API for AI-generated content and evaluation, and Supabase for data storage and semantic retrieval. Preliminary testing indicated that response times were affected by latency in the Gemini API while database and hosting services maintained normal performance. System responsiveness is therefore dependent on the availability and load conditions of external providers, which are outside the researchers' control.

Content generation may occasionally fail when the external AI provider is unavailable or when request limits are reached. Such

failures are transient and are handled through retry mechanisms, but the platform cannot guarantee that every generation request will succeed on the first attempt.

Performance Efficiency is measured under live operating conditions, so ratings reflect the combined response time of the platform and the external services it depends on. Results for this characteristic should therefore be interpreted as end-to-end system performance rather than as a measure of the platform's internal processing alone.

CHAPTER II

INTRODUCTION

This chapter presents the methodology used to design, develop, and evaluate LearnSync AI. This web-based platform is created for Junior High School (JHS) students to support them in their independent learning, with an accompanying Educator Portal that gives teachers direct control over the learning materials their students access. The sections in this chapter detail the system's technical design, development workflows, algorithms, and data structures that allow the platform to operate safely, securely, and efficiently.

The application is designed with two distinct user roles. The Educator uploads and manages the learning materials that students will use, ensuring that all study content has been reviewed and validated by a qualified teacher before it reaches the learning environment. The Educator can also monitor the learning mastery progress of their students through a class-level analytics dashboard. The Student accesses these educator-provided materials and engages with the platform's AI-assisted learning features, including Adaptive Flashcards, Adaptive Test Mode, the Feynman Technique, Free Recall, AI-generated educational games, and a personal analytics dashboard. Bayesian Knowledge Tracing (BKT) continuously estimates each student's mastery and uses the resulting scores to personalize the difficulty of learning activities.

Finally, this chapter describes the system implementation and evaluation procedures. A 2 to 4 week implementation and user evaluation will be conducted

at Pasig Christian Academy with Junior High School students and their assigned educators serving as the study participants. User feedback will be collected using an evaluation instrument based on the ISO/IEC 25010 software quality model to assess the platform's Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The results will be used to determine the system's quality and overall acceptability as an AI-assisted learning platform.

RESEARCH DESIGN

This study uses a developmental research design with a quantitative approach to guide the design, construction, and evaluation of the LearnSync AI web application. In educational technology, developmental research focuses on the systematic creation, testing, and evaluation of instructional tools to ensure they are stable, functional, and effective (Angeles & Cortez, 2023). By using this design, this project transitions from theoretical software planning to a functional, real-world application tested by its target users. The systematic recording of the design and development processes allows the research team to translate educational needs into functional software features while collecting objective numbers to prove the application's overall performance.

The developmental process for LearnSync AI is structured into three distinct phases. The first phase, analysis and design, identifies the study challenges faced by Junior High School students during self-paced study and the

supervisory needs of their educators, then uses these findings to design the user interface layouts of the Student Portal and Educator Portal, map out the study space architecture, establish the BKT routing rules, and draft the PlayMe gamification mechanics. The second phase, development, involves the technical implementation of the web application, including configuring the Supabase relational database with Row-Level Security policies, programming the generative AI parser to extract study terms from educator-uploaded materials, implementing the real-time BKT scoring engine, and building the educator-facing class analytics dashboard. The third phase, evaluation, deploys the finished platform over a 2 to 4 week period at Pasig Christian Academy, where educators will upload learning materials through the Educator Portal while Junior High School students independently use the platform's learning features. Participant experiences and feedback will be collected through a structured evaluation questionnaire to assess quality and user acceptability.

To objectively evaluate the developed system, the study utilizes quantitative data collected through two complementary methods. The first method involves the extraction of system-generated operational logs, specifically from students' interactions with the Adaptive Flashcards and Adaptive Test Mode. Within these modules, the Bayesian Knowledge Tracing (BKT) algorithm continuously computes the probability of learner mastery based on assessment performance. This operational data is utilized to validate the functional accuracy of the adaptive learning mechanism and to verify that the system correctly

personalizes subsequent learning activities based on the algorithm's calculated estimates

The second method is a software quality evaluation based on the ISO/IEC 25010 software quality model. After using the platform, the participants will complete a structured evaluation questionnaire using a four-point Likert scale. The collected responses will be analyzed using descriptive statistics to determine the level of acceptability of LearnSync AI in terms of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The results will be used to assess the overall quality and user acceptability of the developed platform.

ARCHITECTURAL FRAMEWORK

The architectural framework of LearnSync AI follows a dual-portal, three-tier model. The Educator Portal handles material upload, grade-level access control, and mastery monitoring. The Student Portal handles independent study, adaptive assessment, and gamified review. The system is divided into three tiers: Presentation, Application, and Data, with information flowing down when a user performs an action and back up when the system returns a result.

1. **Presentation Tier -** This is the interface layer that both educators and students interact with through their web browser. It has four parts. The

Educator Portal allows the educator to upload learning materials as either plain-text PDF modules or as images of printed pages captured from physical textbooks, select which grade level can access them, and view a real-time analytics overview of individual student mastery levels. The Study Space Hub is where students access educator-uploaded materials, rate flashcards, complete Feynman Technique and Free Recall exercises, configure and take practice tests, and interact with the AI chatbot. The PlayMe Portal captures student interactions during Crossword, Term Matching, and Timed Trivia games. The Dashboard Screen displays the student's personal progress, including daily streaks, mastery curves, and earned badges.

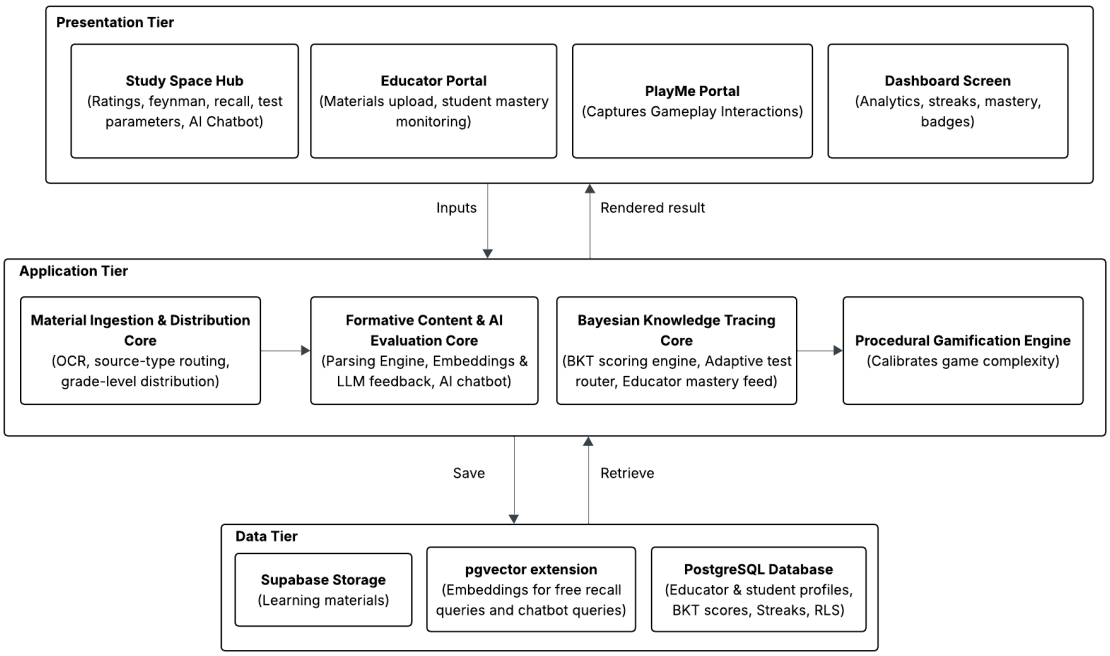
2. **Application Tier -** This is the processing layer. It has four parts. The Material Ingestion and Distribution Core receives educator-uploaded materials and routes them according to source type. Plain-text PDF modules pass directly to the parsing engine. Image-based uploads are first processed through Optical Character Recognition (OCR), which converts the captured page images into machine-readable text before that text is passed to the same parsing engine. This allows lessons that exist only in printed textbooks to enter the same processing pipeline as digital modules. The Core then makes the processed content available to students within the educator's selected grade level. The Formative Content and AI Evaluation Core extracts key terms and concepts from

parsed materials, evaluates student Free Recall and Feynman responses using vector embeddings and LLM feedback, and powers the AI chatbot responses grounded in the uploaded learning resource. The Bayesian Knowledge Tracing (BKT) Core takes flashcard ratings and test answers, calculates per-concept mastery estimates, routes adaptive test difficulty through the Adaptive Test Router, and feeds aggregated mastery data back to the Educator & Student Dashboard. The Procedural Gamification Engine reads the BKT mastery score to calibrate game difficulty to the student's current level.

3. **Data Tier -** This is the storage layer, running on Supabase. Supabase Storage holds the raw source files uploaded by educators, including both plain-text PDF modules and page images captured from printed materials. The pgvector extension stores vector embeddings used for Free Recall similarity checks and AI chatbot retrieval. The PostgreSQL Database holds educator and student profiles, grade-level material access rules, BKT mastery scores, and study streaks, with Row-Level Security (RLS) enforcing data isolation between students and restricting educators to the mastery data of students within their selected grade level.

Figure 1.

Architectural Framework



ALGORITHM

This section details the mathematical and algorithmic foundation of LearnSync AI's cognitive modeling core. To deliver a highly personalized, self-directed learning experience, the platform integrates the classical four-parameter Bayesian Knowledge Tracing (BKT) model. BKT is a probabilistic learner model that has remained a widely used and interpretable approach for tracing student knowledge, and it continues to be validated in modern educational technology research, including through accessible implementations such as the pyBKT framework (Bulut et al., 2023). Within LearnSync AI, the four-parameter update equations are implemented directly in the platform

backend. BKT operates as a dynamic hidden Markov model (HMM) with observable nodes, continuously analyzing a student's sequential interactions to trace their latent knowledge state for any given concept without requiring computationally heavy server resources (Abdelrahman et al., 2023).

1. BKT Mathematical Model - The primary objective of the BKT algorithm is to estimate the probability that a student has mastered a specific concept based on their chronological sequence of correct and incorrect answers. The mathematical framework is governed by four distinct, highly interpretable parameters categorized into two knowledge parameters and two performance parameters (Bulut et al., 2023):

a. **Initial Knowledge State** $P(L_0)$: The prior probability that a student already knows or has mastered a concept before interacting with any formative study modes or assessments on the platform.
b. **Transition Probability** $P(T)$: The learning rate represents the probability that a student shifts from an unmastered (not-known) state to a mastered (known) state after a learning opportunity.
c. **Guess Probability** $P(G)$: The probability that the student has not mastered the concept but is still able to answer correctly.

d. **Slip Probability** $P(S)$: The probability that the student has mastered the concept still answers incorrectly due to carelessness, cognitive fatigue, or a reading error.

As the student interacts with the application, their performance at any given learning opportunity *t* is captured as an observable binary variable (obs_t), where an answer is marked as correct ($obs_t = 1$) or incorrect ($obs_t = 0$). When a new observation is captured, the system executes a two-phase Bayesian update. First, the algorithm applies the evidence update phase to calculate the conditional probability of mastery ($P(L_t \mid obs_t)$ based on the observation:

Formula when the students attempt is correct ($obs_t = 1$) :

EQ (1): $$P(L_t \mid obs_t = 1) = \frac{P(L_t)\,(1-P(S))}{P(L_t)\,(1-P(S)) + (1-P(L_t))\,P(G)}$$

Formula when the students attempt is incorrect ($obs_t = 0$) :

EQ (2): $$P(L_t \mid obs_t = 0) = \frac{P(L_t)\,P(S)}{P(L_t)\,P(S) + (1-P(L_t))\,(1-P(G))}$$

Second, the algorithm performs the transition update phase. This step calculates the projected probability of mastery for the subsequent learning opportunity $P(L_{t+1})$ by factoring in the learning rate $P(T)$ as the student moves to the next card or question, assuming immediate formative feedback is processed:

EQ (3): $$P(L_{t+1}) = P(L_t \mid obs_t) + (1 - P(L_t \mid obs_t))\,P(T)$$

1.1. Default Parameter Initialization - LearnSync AI is newly developed and has no student interaction data, so it cannot calculate its parameters from student responses. Expectation Maximization (EM), the standard method for this, requires a large dataset of prior responses (Bulut et al., 2023; Šarić-Grgić et al., 2024). The system therefore uses fixed values. This does not prevent it from working, as mastery tracking begins at a student's first interaction.

The knowledge tracing literature does not recommend specific numbers for these parameters, since published values are calculated from each study's own data. What it provides instead are limits on acceptable values. These prevent model degeneracy, a condition in which the model behaves illogically, such as lowering a student's mastery after a correct answer. A recent systematic review documents three sets of limits still in use (Šarić-Grgić et al., 2024). As reported there, Corbett and Anderson proposed $P(G) < 0.30$ and $P(S) < 0.10$; Beck and Chang proposed $P(G) + P(S) < 1$; and Baker, Corbett, and Aleven proposed $P(G) < 0.50$ and $P(S) < 0.50$ (all as cited in Šarić-Grgić et al., 2024). The values used in this study are chosen because they fall inside these accepted limits.

The following default parameter values are adopted for all newly created concept nodes upon document upload:

a. $P(L_0) = 0.10$. The starting mastery value is set to 0.10, representing a 10% probability that a student already knows a concept when tracking begins. This conservative value was chosen because the platform receives no prior-knowledge data. Uploaded materials contain no record of what any individual student already knows, and the system performs no diagnostic assessment before tracking begins, so there is no basis on which to set a higher starting value for any particular student. Starting low requires mastery to be established through recorded performance. Of the four parameters, $P(L_0)$ has the shortest-lived effect: because each response multiplies the previous estimate, the starting value has little remaining influence after a few responses.

b. $P(T) = 0.20$. The learning rate is set to 0.20, meaning a student has a 20% chance of moving from not knowing a concept to knowing it at each learning opportunity. The appropriate learning rate depends on what a learning opportunity consists of. Published work has accordingly set different $P(T)$ values for different activity types, adjusting the rate based on the item answered, the type of learning resource used, and the kind of instruction received (Pardos et al., 2013; Badrinath et al., 2021, both as cited in Šarić-Grgić et al., 2024). Conventional learning rate values originate in intelligent tutoring systems, where one learning opportunity is a full guided problem accompanied by instructional feedback. In LearnSync AI, a learning opportunity is a flashcard review or a single test item, completed without instructor feedback, and students carry the full

self-monitoring burden (Luo & Zhou, 2024). A moderately conservative rate is therefore adopted. This study acknowledges that applying a single $P(T)$ to both activity types is a simplification.

c. $P(G)$ = Context-Sensitive (Dynamic). In standard BKT, the guess probability is one fixed value per concept. LearnSync AI instead varies P(G) by assessment mode, question format, and self-reported confidence. This follows established practice. Baker, Corbett, and Aleven (2008, as cited in Šarić-Grgić et al., 2024) showed that letting guess and slip vary between responses produces more accurate student models than fixed per-concept values, and the same review lists question difficulty among the most frequently studied extensions of this kind (Šarić-Grgić et al., 2024). Varying $P(G)$ is mathematically valid because the update formulas accept any value between 0 and 1 for each individual response. The values here are not estimated but calculated: for each format, $P(G)$ is the chance of answering correctly by random guessing, determined by the number of answer choices.

In Adaptive Flashcard Mode, the guess probability varies by the student's self-reported confidence rating. A rating of Easy assigns $P(G_easy) = 0.05$, reflecting high confidence that the correct response was due to genuine knowledge rather than guessing. A rating of Medium assigns $P(G_medium) = 0.25$, accounting for a moderate possibility that the response was aided by partial guessing. A rating of Hard assigns $P(G_hard) = 0.10$ and is recorded as an incorrect response. In the incorrect-answer formula, $P(G)$

appears as the term $(1 - P(G))$ in the denominator, so a lower $P(G)$ produces a larger denominator and a steeper drop in mastery. Setting $P(G_hard) = 0.10$ rather than a higher value therefore ensures that a self-reported failure of recall produces a suitably large decrease.

In Adaptive Test Mode, the guess probability is calibrated per question type based on the mathematical probability of arriving at a correct response through random guessing alone. Because the baseline chance of a correct random response is determined directly by the number of available options, these values follow from the item format itself. The following $P(G)$ values are adopted for each question format:

1. Multiple Choice (4 options): $P(G_mc) = 0.25$. With four answer options and one correct answer, the baseline probability of guessing correctly is 1/4 = 0.25.
2. True or False: $P(G_tf) = 0.50$. With only two possible responses, the baseline probability of guessing correctly is 1/2 = 0.50. This higher guess rate ensures that correct responses on True or False items contribute a substantially smaller mastery boost compared to other question types, reflecting the greater role of chance in binary-choice formats.
3. Identification: $P(G_id) = 0.05$. Open-ended identification items require the student to produce the exact term from memory without any provided

options. The probability of guessing the correct term without prior knowledge is negligibly low.

4. Fill in the Blank: $P(G_fib) = 0.05$. Similar to identification items, fill-in-the-blank questions require free recall of specific terms or phrases without cues. The absence of selectable options makes random guessing highly unlikely, justifying $P(G) = 0.05$.
5. Matching Type (4 terms, 6 definitions): $P(G_match) = 0.17$. For any individual term, the probability of selecting the correct definition from the pool of six through random guessing is 1/6 ≈ 0.17, applied per matched item.

A model is degenerate when its parameters make it behave illogically, such as lowering mastery after a correct answer. The rule preventing this is $P(G) + P(S) < 1$, proposed by Beck and Chang (2007, as cited in Šarić-Grgić et al., 2024) and documented as still in use by the current systematic review of the field.

Every combination used here passes. With $P(S) = 0.15$ throughout, Identification, Fill in the Blank, and Easy-rated flashcards use $P(G) = 0.05$, totalling 0.20; Hard-rated flashcards use 0.10, totalling 0.25; Matching Type uses 0.17, totalling 0.32; Multiple Choice and Medium-rated flashcards use 0.25, totalling 0.40; and True or False uses 0.50, totalling 0.65. Every total is below 1.

Seven of the eight formats also pass both the stricter $P(G) < 0.30$ of Corbett and Anderson and the looser $P(G) < 0.50$ of Baker, Corbett, and

Aleven (2008; both as cited in Šarić-Grgić et al., 2024). True or False is the only exception, which cannot be avoided with a two-choice format where random guessing succeeds half the time. The practical result is that a correct True or False answer raises mastery far less than other formats, correctly reflecting how little it proves.

d. $P(S) = 0.15$. The slip probability is set to 0.15, meaning a student who genuinely knows a concept still answers incorrectly 15% of the time through carelessness, misreading, or fatigue. Checked against the three sets of limits documented in the current systematic review of the field (Šarić-Grgić et al., 2024), this value passes the degeneracy rule $P(G) + P(S) < 1$ of Beck and Chang for every question format used, and passes the limit $P(S) < 0.50$ of Baker, Corbett, and Aleven. It exceeds only the strictest limit, $P(S) < 0.10$ of Corbett and Anderson (1995; all as cited in Šarić-Grgić et al., 2024). This is deliberate. That limit was set for tutoring systems with real-time teacher supervision and immediate correction, whereas LearnSync AI is a self-directed platform where students work alone and are more prone to lapses in attention (Luo & Zhou, 2024). A value of 0.15 remains well below 0.50 and keeps the model meaningful.

How the fixed parameters behave. The model contains two kinds of values. $P(L)$, the mastery estimate, changes after every student response. The four parameters $P(L_0)$, $P(T)$, $P(S)$, and $P(G)$ never change. They control how each update works and remain constant for as long as the system runs. This

distinction is stated because the two are easily conflated. The Bayesian updating mechanism refines the mastery estimate; it does not refine the parameters. The parameters are accordingly described in this study as selected configuration values operating within published limits, rather than as values the system converges toward on its own. One effect of the update mechanism is worth noting: because each response multiplies the previous estimate, the influence of the starting value $P(L_0)$ fades quickly, and after a few responses the mastery estimate is driven almost entirely by demonstrated performance. Fixed parameters are standard for BKT implementations operating without historical training data, which is why the literature treats calculating parameters from data as the preferred approach wherever sufficient data exists (Šarić-Grgić et al., 2024). This study therefore verifies that every adopted value falls inside the limits published in the literature, and identifies recalculating the parameters from collected interaction data after deployment as future work.

1.2. Step-by-Step Computation of Student Mastery - The following trace shows how the system updates a student's mastery of a single concept across three consecutive multiple-choice attempts using the default parameters: $P(L_0) = 0.10$, $P(T) = 0.20$, $P(S) = 0.15$, $P(G) = 0.25$. Each attempt applies the evidence update first, then the transition update.

Attempt 1 - Correct $obs_t = 1$, prior $P(L_0) = 0.10$:

$$P(L_0 \mid obs_t = 1) = \frac{(0.10)(1-0.15)}{(0.10)(1-0.15)+(1-0.10)(0.25)} = 0.27$$

$$P(L_{0+1}) = 0.27 + (1 - 0.27)\,0.20 = 0.42$$

Attempt 2 - Correct $obs_t = 1$, prior $P(L_1) = 0.42$:

$$P(L_1 \mid obs_t = 1) = \frac{(0.42)(1-0.15)}{(0.42)(1-0.15)+(1-0.42)(0.25)} = 0.71$$

$$P(L_{1+1}) = 0.71 + (1 - 0.71)\,0.20 = 0.77$$

Attempt 3 - Incorrect $obs_t = 0$, prior $P(L_2) = 0.77$:

$$P(L_2 \mid obs_t = 0) = \frac{(0.77)(0.15)}{(0.77)(0.15)+(1-0.77)(1-0.25)} = 0.40$$

$$P(L_{2+1}) = 0.40 + (1 - 0.40)\,0.20 = 0.52$$

Two consecutive correct answers raised the student's mastery from 0.10 to 0.77, while a single subsequent incorrect answer reduced it to 0.52, a notable but controlled drop, demonstrating how the slip parameter cushions careless mistakes without erasing accumulated progress

2. Algorithmic Implementation in Flashcard Mode - Within the study space, the spaced repetition manages how the flashcards behave while BKT calculates the student's mastery of a concept. These flashcards can be made by the student or generated by the AI using the uploaded files. When a student opens a flashcard session, the system loads their current progress from the database. If it

is a brand-new card set, the system initializes their score to the default prior $P(L_0) = 0.10$. During study, the student flips a card to reveal its definition and rates how well they remembered it by choosing Easy, Medium, or Hard.

Each rating maps to a binary observation and to the confidence-based guess probability established in Section 1.1, and it simultaneously sets the spaced-repetition interval at which the card will reappear. Choosing Easy means the student remembered the card easily; the system records a correct answer ($obs = 1$), applies the low-confidence guess rate $P(G_easy)$ so that the mastery boost reflects genuine knowledge rather than luck, and schedules the card to reappear in seven days. This longer interval corresponds to a later review tier of the Leitner spaced-repetition system, which prescribes progressively longer intervals for well-recalled items to optimize long-term retention while minimizing unnecessary review (Durrani et al., 2024).

Choosing Medium means the student remembered the card but had to think about it; the system again records a correct answer ($obs = 1$) but applies the higher medium-confidence guess rate $P(G_medium)$, yielding a smaller and more cautious mastery boost, and schedules the card to reappear in three days. This interval aligns with an intermediate review tier of the Leitner system, ensuring moderately recalled items are revisited sooner than well-recalled items to reinforce retention before significant forgetting occurs (Durrani et al., 2024).

Finally, choosing Hard means the student forgot the card; the system records an incorrect answer ($obs = 0$), applies the hard-confidence guess rate $P(G_hard)$ from Section 1.1, and schedules the card to reappear in one day for immediate re-exposure. This shortest interval corresponds to the first review tier of the Leitner system, which prioritizes daily review for unrecalled or difficult items to maximize re-learning opportunities (Durrani et al., 2024). In every case the rating is sent directly to the BKT math engine, which applies the two-phase update from Section 1.1 to produce the student's new progress percentage $P(L_{+1})$ and saves it to the database. This hybrid setup gives students personalized review times while the BKT engine accurately updates their knowledge statistics behind the scenes.

3. Algorithmic Implementation in Adaptive test Mode - Within the adaptive test mode, the Bayesian Knowledge Tracing model serves as the main engine for calibrating the difficulty of generated assessment sessions. When entering Test Mode, the student selects one or more custom question formats, such as multiple choice, identification, matching type, fill in the blank, or true or false. As the student completes their attempts during active testing, their performance is graded as a binary value (1 for correct or 0 for incorrect), and the BKT update pipeline calculates their dynamic mastery score on a per-question basis. Each attempt is evaluated using the question-type-specific guess probability defined in Section 1.1 together with the fixed slip probability $P(S) = 0.15$, so that the mastery gain from a correct answer is automatically scaled to how much chance

the format allows. This real-time calculated score updates the student's concept-specific history in the central database.

When the student subsequently requests the generation of a new practice test session on the same topic, the system queries this historical BKT mastery probability to scale the conceptual difficulty of the newly generated items. The adaptive test generation engine utilizes specific mastery thresholds to adjust the generation prompt instructions sent to the large language model.

If the current running mastery probability is high, specifically at or above 0.85, the AI model is instructed to generate advanced-tier questions focusing on analytical scenarios, synthesis, and edge cases within the student's selected formats. If the mastery probability lies in the intermediate range between 0.50 and 0.85, the prompt parameters dictate intermediate-tier questions emphasizing conceptual comparisons, multi-concept recall, and practical applications. If the running mastery sits in the foundational range below 0.50, the generation instructions shift to produce simpler, highly supportive retrieval questions focusing on direct definitions and basic conceptual associations. This session-based adaptation allows students to retain complete control over their preferred test layouts while the system continuously adjusts the cognitive demands of the content to match their developmental needs.

4. Document Mastery and Gamification Calibration - While Bayesian Knowledge Tracing tracks individual concepts as isolated, high-precision

knowledge components in the database, the system must provide a consolidated progress index for end-users. When a teacher of a student uploads a study document, the document ingestion engine parses and extracts a set of N distinct concept nodes. To render a high-readability overview on the student's dashboard, the application layer dynamically aggregates these independent probabilities using an arithmetic mean. The overall mastery rate M_doc for an uploaded document containing N concepts is calculated at any given timestamp using the most current running BKT metrics fetched from the database:

$$\textbf{EQ (4):}\ M_{doc} = \frac{\Sigma\, P(L_{current,\,i})}{N}$$

Where M_doc represents the overall document mastery percentage displayed as a progress bar on the user dashboard; N represents the total number of distinct concepts extracted from the uploaded document file; and $P(L_current,\ i)$ represents the real-time, running BKT mastery probability of the i-th concept stored under the student's profile in the database. Each concept's probability is calculated through the sequential, chronological application of the core BKT equations detailed in the BKT Mathematical Model. At the initial state (before any student interactions, where t = 0), the concept's probability is initialized to the baseline prior $P(L_0) = 0.10$. For each subsequent practice attempt, the system captures the binary response, calculates the intermediate conditional probability using the corresponding evidence formula, and executes the transition update to produce the updated probability state. The active value referenced in this aggregation is the most recently computed and saved $P(L_{+1})$

stored within the student's concept profile, ensuring the aggregate overview dynamically adapts to their live learning progress.

This aggregated score is directly utilized by the system's gamification module to calibrate the operational difficulty of generated games (crosswords, term matching, and timed trivia). When a game session is initialized, the system retrieves the current M_{doc} value from the database and applies it as a baseline parameter to the generative AI prompt. For example, if a student's document mastery sits at exactly 50%, the game engine utilizes this value to restrict the vocabulary complexity, clue difficulty, and trivia response timers to align exactly with that 50% knowledge threshold. This integration protects students from cognitive frustration while keeping gamified learning tightly synchronized with their cognitive progress.

5. Evaluation in Free Recall - Within the study spaces, the Free Recall tab lets students type out everything they can remember about a lesson without looking at their notes. To grade this automatically, the system uses a method called Cosine Similarity which is calculated natively inside the database using the Supabase pgvector extension. This database-level calculation removes the need for complex mathematical scripts in the application's backend code by computing vector geometry directly on the cloud server.

When a student uploads their study materials, the application breaks the document into small pieces and converts those pieces into lists of numbers

called high-dimensional vector embeddings that capture their exact meaning. When the student types out their recall summary, the system converts their typed text into a vector too.

Instead of running slow calculations on the backend application layer, the backend simply sends the student's vector to Supabase. The database natively calculates the mathematical angle between the student's typed vector $\vec{A}$ and the lesson's stored vector $\vec{B}$ using the Cosine Distance Operator <=>, where the final Cosine Similarity is found through the simple subtraction formula:

$$\textbf{EQ (5):}\ Cosine\ Similarity\ =\ 1\ -\ (A\ <=>\ B)$$

If the database returns a score close to 1, it means the student's written ideas point in the same directional meaning as the lesson. If the similarity score falls below a set quality threshold (such as 0.75), the system identifies that a gap exists. The system then compares the student's typed text directly against the lesson's key glossary terms to spot exactly which keywords were missing, displaying this feedback immediately to the student so they know what to review.

6. Feedback generation in Feynman Technique - The Feynman Technique helps students learn by explaining hard topics in simple, everyday language, as if they are teaching a 10-year-old child. To evaluate how well a student does this, the application guides a large language model (LLM) through a three-step evaluation pipeline.

The first step is the Jargon Check. The AI scans the student's written explanation and compares it against the main glossary list extracted during the document upload phase. If the student uses highly complex terms (like "photosynthesis" or "electromagnetism") without breaking them down simply, the AI flags those words and tells the student they need to explain them in simpler terms.

The second step is Analogy Verification. The AI checks if the student used a helpful comparison or everyday story (an analogy) to make the idea clear. For example, when explaining electrical circuits, the AI checks if the student compared it to water flowing through pipes to make the concept easier to grasp.

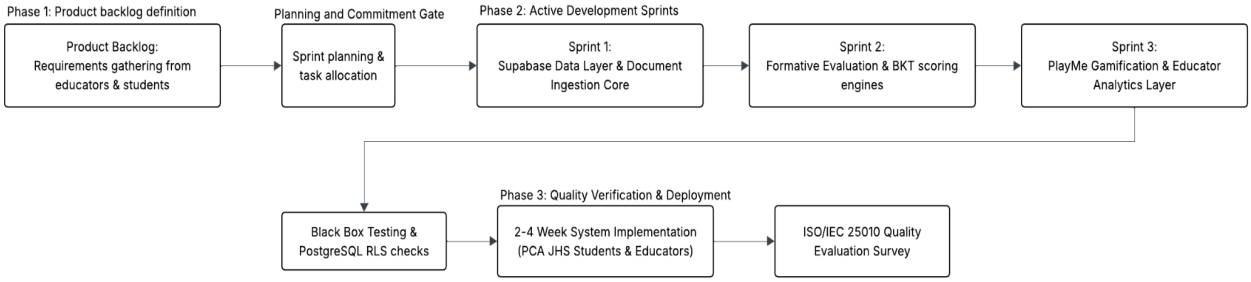
The third and final step is Finding the Gaps. The AI compares the student's simple explanation against the concept's true definition stored in the system to check whether anything is factually wrong, missing, or incomplete. Based on this comparison, it produces an accuracy score from 0 to 100 and a matching verdict of Correct, Needs Improvement, or Incorrect. The AI does not just hand back a grade. It also writes short, encouraging feedback that points out what the student got right, names any factual errors directly, and tells them specifically how to improve so they can close their own learning gaps.

SOFTWARE ENGINEERING PARADIGM

This study utilizes the Agile Scrum framework to guide the design, development, and implementation of LearnSync AI. Since the platform includes several interconnected features that require continuous testing and improvement, a flexible software development methodology is needed. Agile Scrum supports iterative development through short sprint cycles, allowing developers to continuously build, test, and refine the system while responding to changing requirements and user feedback (Hodson et al., 2024). Using Agile Scrum also enables the researchers to identify and resolve issues early in the development process, resulting in better software quality and reduced development risks (Strode et al., 2022). This approach ensures that LearnSync AI is developed efficiently while meeting both user needs and system requirements.

Figure 2.

Agile SDLC



The execution of the Agile Scrum framework across the development lifecycle is structured into the following sequential phases:

1. **Phase 1: Product Backlog** - The research team establishes the master list of software requirements from the perspective of two user roles: the Educator and the Student. The Educator must be able to upload their learning materials, select which JHS grade level can access the uploaded materials, and monitor student mastery through a class-level analytics dashboard. The Student must be able to access educator-uploaded materials and engage with four self-paced study options (adaptive flashcards, Free Recall with AI feedback, Feynman Technique explanations, and an AI chatbot), generate customized adaptive practice tests of various formats (multiple-choice, matching, fill-in-the-blank, identification, and true/false) that scale in difficulty based on their BKT mastery state, and access a gamified review engine (crosswords, matching cards, and timed trivia) that adjusts difficulty using BKT scores.

2. **Planning and Commitment Stage** - The team conducts sprint planning sessions to break down backlog specifications into modular engineering tasks, estimate complexities, designate priorities, allocate tasks, and establish a sprint calendar. This ensures development cycles remain focused, preventing scope creep and prioritizing high-risk mathematical models and data security layers before active coding begins.

3. **Phase 2: Active Development Sprints -** The core coding lifecycle, structured into three iterative sprints:

    a. **Sprint 1: Supabase Data Layer & Document Ingestion Core** - Configures PostgreSQL tables on Supabase for educator profiles, student profiles, grade-level material access rules, and learning materials and embeddings records. Enforces Row-Level Security (RLS) policies to isolate each student's data under their authenticated user ID and restrict educators to mastery data within their selected grade level. Implements LLM-based text extraction scripts to parse educator-uploaded materials, extract concepts and definitions, and generate semantic vectors stored directly in the database.

    b. **Sprint 2: Formative Evaluation & BKT Scoring Engines** - Programs the four-parameter BKT scoring calculations to track student mastery after each attempt (0 for incorrect, 1 for correct) and develops the adaptive test routing algorithm that adjusts question difficulty based on running mastery metrics. Establishes vector-based semantic evaluation for Free Recall similarity checks, integrates LLM prompt templates for Feynman explanations and

chatbot queries, and logs activity timestamps to the Activity and Streak Logs database.

c. **Sprint 3: PlayMe Gamification & Educator Analytics Layer**: Codes the backtracking layout algorithm for crossword grids, trivia questions, and vocabulary matches using parsed glossaries and BKT scores as difficulty references. Builds the gamification and streak engine to validate badge criteria and manage daily streaks from a unified activity log. Programs the educator analytics dashboard to aggregate student mastery data at the concept and document level into a real-time grade-level monitoring interface. Compiles the backend API gateways with a responsive, dual-portal web frontend serving both Educator and Student interfaces.

4. **Phase 3: Quality Verification & Development** - The completed application undergoes Black Box Testing and PostgreSQL RLS checks to verify that data isolation operates correctly for both student-to-student and educator access boundaries. The application is then launched for a 2 to 4 week live implementation at Pasig Christian Academy, where educators upload materials through the Educator Portal while JHS students independently use the platform to study their curriculum materials. At the conclusion, a structured ISO/IEC 25010 quality evaluation survey is

administered using a 4-point Likert scale to measure Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability

DATA FLOW DIAGRAM

This section presents the Data Flow Diagrams (DFDs) for LearnSync AI. A DFD shows how information moves through the system, from the moment an educator uploads a learning material to the point where the system returns adaptive tests, feedback, and progress updates to students and mastery reports to educators. The Level 0 diagram gives a single high-level view of the whole system, and the Level 1 diagram breaks that system down into its main processes and data stores.

1. **Level 0: Context Diagram -** The context diagram represents the entire application as a single process, 0.0 LearnSync AI, with two external entities: the Educator and the Student.

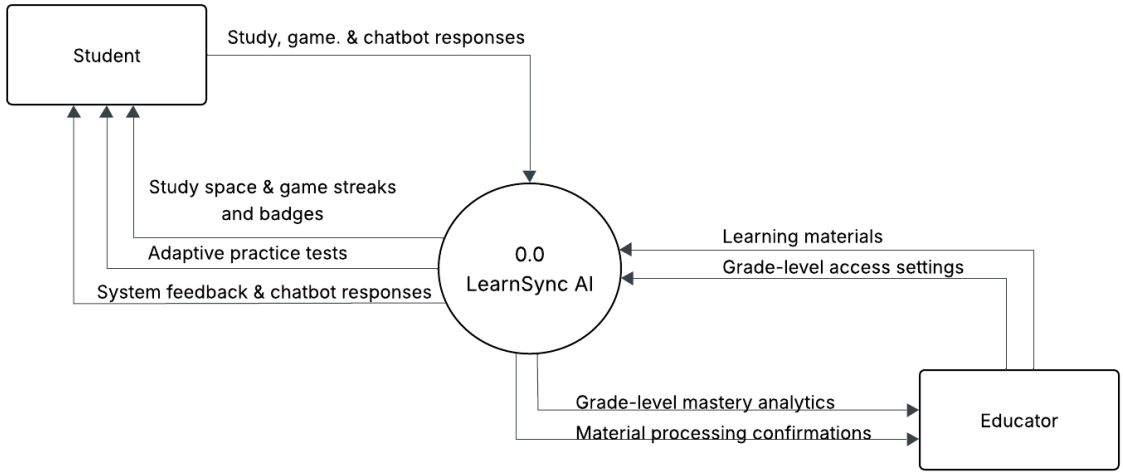
   The Educator sends Learning materials (plain-text PDF modules or images of printed pages) and Grade-level access settings (specifying which grade level can access the uploaded materials) into the system. The system returns Grade-level mastery analytics (real-time student mastery progress reports) and Material processing confirmations (status

responses confirming whether uploads were successfully parsed and made available).

The Student sends Study, game & chatbot responses (flashcard ratings, test answers, game interactions, and chatbot queries) into the system. The system returns three outputs: Study space & game streaks and badges (daily streaks & earned badges), Adaptive practice tests (quizzes scaled to the student's mastery level), and System feedback & chatbot responses (AI-generated study guidance and chatbot replies).

Figure 3.

Level 0 Context Diagram



2. **Level 1: Date flow diagram** - The Level 1 diagram contains 5 processes, 3 data stores (D1: Study Spaces, Learning Materials and Embeddings; D2: Student Profiles and Mastery Track; D3: Activity and Streak Logs), and 2 external entities (Educator, Student).

    Process 1.0 Material Upload and Distribution Control receives Learning materials and Grade-level access settings from the Educator, then stores the raw files and access configuration in D1 through the Raw files & materials access rules flow. When a Student opens their study space, the process checks D1 for matching access rules and returns the Authorized learning materials for that grade level. Process 1.0 also reads aggregated mastery data from D2 through the Grade-level mastery analytics flow and delivers it to the Educator for student progress monitoring.

    Process 2.0 Document Parsing and Semantic Ingestion retrieves raw files from D1 through the Raw uploaded documents flow, extracts keywords, and converts the text into vector embeddings. It writes the results back to D1 through the Parsed text, keywords and vectors flow and sends a Material processing confirmation to the Educator.

    Process 3.0 Formative Study and AI Evaluation handles flashcards, the Feynman Technique, free recall, and the AI chatbot. It receives Formative responses & chatbot queries from the Student, reads Parsed Content from D1, and returns System feedback & chatbot responses.

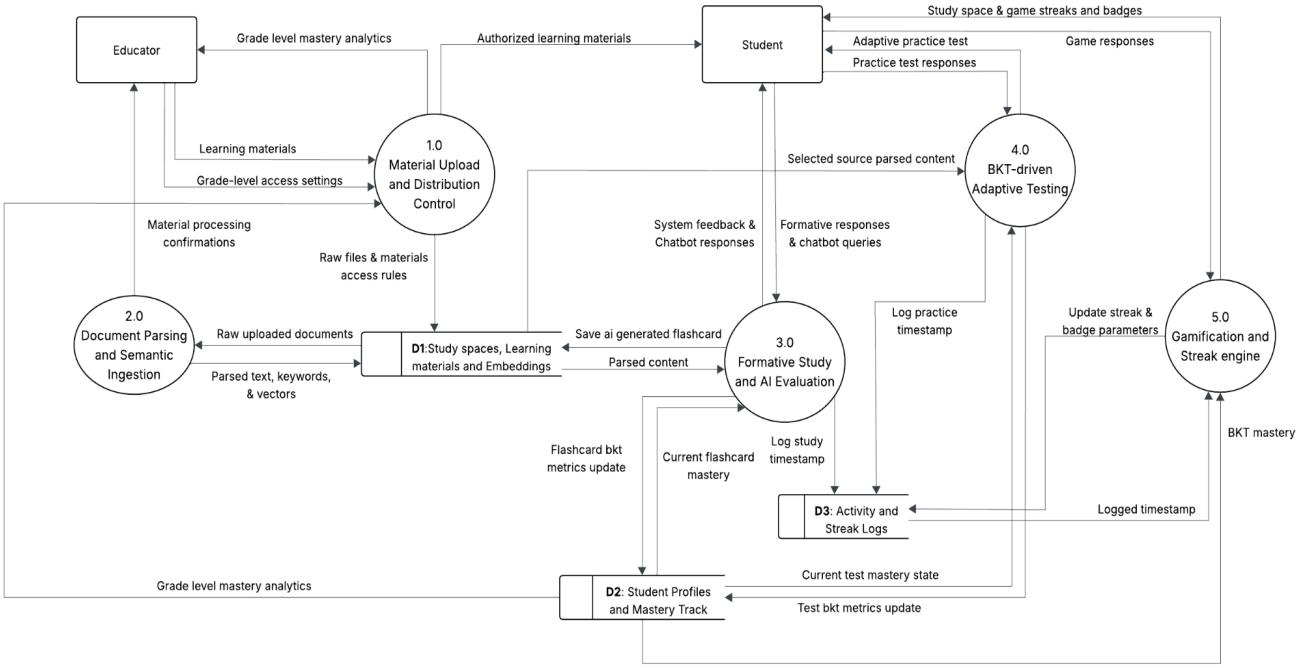
AI-generated flashcards are saved back to D1 through the Save AI-generated Flashcards flow. For mastery tracking, the process reads the student's Current flashcard mastery from D2, applies the BKT calculation based on the flashcard rating, and writes the updated score back to D2 through the Flashcard BKT metrics update flow. It also logs each session to D3 through the Log study timestamp flow.

Process 4.0 BKT-driven Adaptive Testing receives Practice test responses from the Student, reads the Selected source parsed content from D1 and the Current test mastery state from D2, applies BKT to estimate mastery, and returns Adaptive practice tests matched to the student's level. Updated scores are written to D2 through the Test BKT metrics update flow, and session times are logged to D3 through the Log practice timestamp flow.

Process 5.0 Gamification and Streak Engine receives Game responses from the Student, reads the document-level BKT mastery score (M_doc) from D2 to calibrate game difficulty, and reads the Logged timestamp from D3 to check daily study activity. It updates streak and badge records in D3 through the Update streak & badge parameters flow and returns Study space & game streaks and badges to the Student.

Figure 4.

Level 1 Main Process



TOOL TO TEST FOR ACCURACY

To ensure the reliability and correctness of LearnSync AI, the study employs a multi-layered validation strategy that evaluates both the algorithmic implementation and the functional behavior of the system. Since LearnSync AI does not utilize historical student interaction data, model training, or data-driven parameter estimation, the evaluation does not employ traditional machine learning metrics such as Confusion Matrix, Precision, Recall, F1-Score, or Mean Absolute Error. Instead, the study adopts a software engineering and implementation verification approach that validates whether the system's core modules, the Bayesian Knowledge Tracing (BKT) engine and the Generative AI

components, operate correctly according to their specified mathematical formulas and intended functional behavior.

1. **BKT Algorithm Implementation Verification**

The BKT module is evaluated through deterministic algorithm verification, confirming whether the system's programmatic implementation produces mastery estimates that are mathematically identical to hand-computed expected values. Since the study utilizes predefined default parameters from established educational data mining literature rather than parameters learned through Expectation Maximization, the objective is not to assess whether BKT accurately predicts real-world mastery, but to verify that the implemented code faithfully executes the standard four-parameter update equations without computational errors or logical defects.

The researchers design structured test cases, each consisting of a known initial mastery state, a specific sequence of binary responses (correct or incorrect), the default parameters $P(L_0) = 0.10$, $P(T) = 0.20$, $P(S) = 0.15$, and the context-sensitive P(G) values corresponding to the interaction mode, along with the expected mastery output manually computed using the BKT evidence and transition update formulas. These test cases cover consecutive correct responses, consecutive incorrect responses, alternating responses, and mixed sequences across Adaptive

Flashcard ratings (Easy, Medium, Hard) and Adaptive Test Mode question types (multiple choice, true or false, identification, fill in the blank, and matching type). Each test case is executed in the live application, and the system-generated mastery value is compared against the manually computed expected value.

Implementation correctness is quantified using the Implementation Correctness Rate (ICR):

$$\textbf{EQ (6):}\ ICR = \frac{Number\ of\ Passed\ Test\ Cases}{Total\ Number\ of\ Test\ Cases}\ X\ 100$$

A test case is considered passed if the absolute difference between the system-generated and manually computed values falls within a tolerance of ± 0.01, accounting for floating-point arithmetic variations. An ICR of 100% indicates full and correct implementation. Any failed test case is documented with its expected value, actual output, and computed deviation for traceability.

This verification validates the correctness of the algorithm's implementation, not the predictive validity of the BKT model itself. The study confirms that given input parameters and observed responses, the system produces mastery estimates mathematically consistent with the standard BKT formulas (Bulut et al., 2023).

2. Generative AI Functional Verification

The Generative AI components, AI-generated flashcards, adaptive test questions, educational game content, chatbot responses, Feynman Technique feedback, and Free Recall evaluation feedback are evaluated through functional verification rather than model-level benchmarking. Since the study does not fine-tune or retrain the underlying large language model, the evaluation does not assess the AI model's internal accuracy. Instead, it verifies that outputs are contextually relevant, factually consistent with the uploaded learning resource, aligned with the intended BKT-driven difficulty level, and appropriate for the specified interaction mode.

The researchers upload a controlled learning resource with known content and systematically trigger each AI-powered feature. Each generated output is evaluated against four criteria:

Contextual Relevance assesses whether the generated content (flashcard terms and definitions, quiz questions and answers, game clues, chatbot responses, and feedback text) directly pertains to the topics, concepts, and terminology present in the uploaded learning resource. Outputs that reference information outside the scope of the uploaded material or introduce unrelated content are flagged as contextually irrelevant.

Factual Correctness assesses whether the generated content is factually accurate compared to the source material, including quiz answer keys, flashcard definitions, and chatbot responses.

Difficulty Alignment assesses whether the content reflects the appropriate difficulty tier based on the student's current BKT mastery level: foundational (below 0.50), intermediate (0.50–0.85), or advanced (at or above 0.85). This confirms that the adaptive calibration pipeline correctly passes the mastery parameter to the generative AI prompt.

Content Consistency assesses whether the AI produces stable, non-contradictory outputs across multiple generations for the same resource and mastery level. Minor phrasing variations are expected, but core factual content, terminology, and difficulty level should remain consistent.

Results are reported descriptively, documenting the total outputs evaluated, the number and percentage satisfying all criteria, and any notable deviation patterns.

3. Black-Box Functional Testing

The study employs Black-Box Functional Testing to validate the overall behavior of LearnSync AI as an integrated software system. Black-box testing evaluates external behavior based on specified functional requirements without examining internal source code or database structure.

The researchers design test cases corresponding to the platform's core functional requirements for both user roles. Educator-side test cases include PDF document upload and content extraction & grade-level access configuration for uploaded materials, and the class-level mastery analytics dashboard. Student-side test cases include authentication, access to educator-uploaded learning materials, flashcard generation and spaced repetition scheduling, adaptive test session generation and scoring, educational game generation and difficulty calibration, BKT mastery tracking and dashboard visualization, Free Recall submission and cosine similarity evaluation, Feynman Technique submission and AI feedback generation, AI chatbot interaction, and learning analytics display. Each test case specifies a precondition, an input action, and an expected output. The test is executed on the live deployed application, and results are documented in a structured log recording the feature tested, the expected result, the actual result, and the pass or fail status.

4. **ISO/IEC 25010 Software Quality Model Survey**

To evaluate overall software quality from the end user's perspective, the study employs the ISO/IEC 25010 Software Quality Model. The platform is evaluated across the eight quality characteristics defined by the standard: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

A structured survey questionnaire aligned with the ISO/IEC 25010 characteristics is administered to the participants following the 2 to 4 week implementation period. The questionnaire utilizes a Likert scale to capture perceived quality across the eight characteristics. Responses are analyzed using descriptive statistics, including mean scores and standard deviations, to derive an overall quality assessment per characteristic. These results provide empirical evidence of the platform's perceived quality and acceptability, complementing the technical verification results from the BKT algorithm verification, Generative AI functional verification, and black-box functional testing.

POPULATION AND SAMPLING

The target population of this study consists of Junior High School (JHS) students (Grades 7-10) of Pasig Christian Academy for the Academic Year 2026-2027. As the intended end-users of LearnSync AI, these students will participate in the implementation and evaluation of the proposed system. Junior

High School students are the target population as they are in a stage of developing independent learning habits, self-regulated learning skills, and academic readiness, making them suitable participants for evaluating the effectiveness of the proposed platform.

The study will employ Proportionate Stratified Random Sampling to select 50 Junior High School students from the identified accessible population of 100 students. Respondents will first be grouped according to their respective grade levels (Grades 7–10). From each grade level, 50% of the students will be selected, with the number of respondents determined proportionally based on the population of each grade level, as presented in Table 1. Within each grade level, the required respondents will be selected through simple random sampling. This sampling technique ensures that each grade level is proportionally represented in the sample while minimizing selection bias.

Table 1.

Population and Sample Distribution

Grade Level	Population (N)	Sample (n)
Grade 7	18	9
Grade 8	29	15
Grade 9	34	17
Grade 10	19	9
Total	100	50

The sample size for each grade level was determined using the proportionate allocation formula:

$$\textbf{EQ (7):}\ n_h = \frac{N_h}{N}\ x\ n$$

Where:

- n_h = Sample size for each grade level
- N_h = Population of each grade level
- N = Total Population
- n = Total Sample Size

For the educator respondents, two educators responsible for the subject areas identified during the requirements analysis as difficult among the participating students will be included. Since only two educators meet the identified criteria, total enumeration will be employed, wherein both eligible educators will be invited to evaluate the educator functionalities of LearnSync AI.

Although the respondents will be selected from different grade levels, the study does not intend to compare learning outcomes across Grades 7, 8, 9, and 10. Instead, all selected participants will be analyzed as a single study population for the implementation and evaluation of LearnSync AI. During the implementation period, the respondents will utilize the platform's core features,

including Adaptive Flashcards, Adaptive Test Mode, AI-assisted study strategies, AI-generated educational games, the AI-powered learning assistant, and learning analytics. The analysis will focus on changes in students' learning performance, measured through the difference between their pre-test and parallel post-test scores, rather than differences in learning outcomes among grade-level groups.

EVALUATION INSTRUMENT

The primary evaluation instruments used in this study consist of system-generated assessment tools, backend learning analytics, and researcher-adapted questionnaires designed to evaluate LearnSync AI in terms of system effectiveness, software quality, and user acceptability. The evaluation framework consists of four components. First, system-generated parallel pre-tests and post-tests are utilized as primary assessment instruments to measure changes in students' learning performance and concept mastery, while the platform's adaptive features personalize the learning activities based on each student's estimated mastery. Second, the platform automatically extracts system logs to record the students' calculated mastery probabilities and learning progression using the Bayesian Knowledge Tracing (BKT) engine throughout the implementation period. Third, Junior High School students and educators evaluate the acceptability of LearnSync AI using questionnaires based on the ISO/IEC 25010 Software Quality Model. Finally, qualitative feedback is collected to gather participants' comments, observations, and recommendations for future

system improvement. The ISO/IEC 25010 questionnaire evaluates the platform using the applicable software quality characteristics of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Students evaluate the student-facing functionalities of LearnSync AI, while educators evaluate only the functionalities accessible through the educator module, including learning resource upload and management, grade-level assignment of learning resources, and student learning statistics.

- **Quantitative Instrument:** The quantitative instrument consists of structured close-ended statements designed to evaluate the quality and acceptability of LearnSync AI. Respondents rate each statement using a 4-point Likert scale, which eliminates the neutral option to encourage a more definitive evaluation of the system:

    - **4 – Strongly Agree (SA) / Excellent:** The evaluated functionality consistently performs as intended and fully satisfies the expected quality standards.
    - **3 – Agree (A) / Good:** The evaluated functionality performs satisfactorily and meets the expected quality standards, with only minor issues that do not significantly affect its overall performance or usability.

- **2 – Disagree (D) / Fair:** The evaluated functionality partially meets the expected quality standards but exhibits noticeable limitations or inconsistencies that affect its performance or usability.
- **1 – Strongly Disagree (SD) / Poor:** The evaluated functionality fails to meet the expected quality standards and demonstrates significant deficiencies that negatively affect its performance, usability, or intended operation.

- **Qualitative Feedback:** The final section consists of open-ended questions that allow both students and educators to provide comments, observations, suggestions, and recommendations regarding LearnSync AI. This feedback enables respondents to describe their experiences using the platform, identify usability or functional concerns, and recommend possible improvements that may not be adequately captured through the quantitative evaluation.

DATA COLLECTION METHOD

Data for this study will be collected through four phases: requirement analysis, pre-implementation, system implementation, and post-implementation evaluation. These phases ensure that the identified needs of Pasig Christian

Academy are translated into the design, development, implementation, and evaluation of LearnSync AI.

During the requirement analysis phase, the researchers will conduct interviews and questionnaires with the school principal, educators, and Junior High School students of Pasig Christian Academy to identify the current independent learning process, existing challenges, and the functional and non-functional requirements of the proposed system. The information gathered during this phase will serve as the basis for the design and development of LearnSync AI.

During the pre-implementation phase, the researchers will obtain permission from the school administration. The objectives, procedures, and duration of the study will be explained, and participants will receive an orientation on how to access and use LearnSync AI.

During the implementation phase, 50 selected Junior High School students will use LearnSync AI for two to four weeks as part of their independent learning outside regular classroom instruction. The implementation will initially focus on subject areas identified through the requirements analysis as difficult for students in each participating grade level and appropriate for the current concept-based learning capabilities of LearnSync AI. This subject selection is intended to provide the initial learning content for system evaluation and does not restrict the platform to a specific subject area or prevent its future application to other subjects.

After accessing the learning materials, students will complete a system-generated standard pre-test. The students will utilize the platform's core features, including Adaptive Flashcards, Adaptive Test Mode, the AI Learning Assistant, AI-assisted study strategies through the Feynman Technique and Free Recall, AI-generated educational games, and learning analytics.

After completing the designated learning module, students will take a parallel post-test. Throughout the implementation, the system will automatically record each student's pre-test and post-test scores, study activity, and Bayesian Knowledge Tracing (BKT) mastery estimates. The researchers will monitor the implementation and provide technical assistance whenever necessary.

During the post-implementation phase, Junior High School students and educators will complete the ISO/IEC 25010 Software Quality Evaluation Questionnaire through Google Forms. Students will evaluate the student-facing functionalities of LearnSync AI, while educators will evaluate only the functionalities available through the educator module. An open-ended questionnaire will also be administered to gather comments, suggestions, and recommendations for future system improvements.

The responses collected through Google Forms will be automatically recorded for statistical analysis of the system's software quality and support for independent learning. Concurrently, the system-generated pre-test and post-test scores will be analyzed to determine whether there is a significant change in students' learning performance following the implementation period. Meanwhile,

the BKT records extracted from LearnSync AI will be used to examine the operation of the platform's adaptive personalization throughout the implementation period.

ETHICAL CONSIDERATIONS

This study will be conducted in accordance with established ethical research principles to protect the rights, privacy, and welfare of all participants. Prior to the implementation of LearnSync AI, permission will be obtained from the administration of Pasig Christian Academy. Participants will be informed of the purpose, procedures, duration, and nature of their participation in the study. Participation will be voluntary, and participants may decline or discontinue their participation without penalty.

The study will comply with the Data Privacy Act of 2012 (Republic Act No. 10173) by collecting only the information necessary for system implementation and research. Personal information, academic records, and system-generated learning data, including Bayesian Knowledge Tracing (BKT) mastery records, will be treated with confidentiality and used solely for academic research and system evaluation. No personally identifiable information will be disclosed in reports, presentations, or publications.

To maintain academic integrity, only authenticated educators will be authorized to upload learning resources used by the platform. Access to learning analytics and personal learning records will be restricted to the respective

student and authorized educators. All collected information will be handled securely and will not be shared with unauthorized third parties.

SUMMARY

This chapter presented the research methodology used in the development and evaluation of LearnSync AI, an AI-assisted learning platform designed to support the independent learning of Junior High School students with an accompanying Educator Portal for material management and student mastery monitoring. It discussed the research design, system architecture, software engineering methodology, research locale, population and sampling, evaluation instrument, data collection procedures, system validation methods, statistical treatment of data, and ethical considerations.

The study employed a developmental research design using quantitative methods to support the analysis, design, development, implementation, and evaluation of the proposed system. LearnSync AI was developed using the Agile Scrum software development methodology and integrates the classical Bayesian Knowledge Tracing (BKT) model to estimate students' mastery levels. Educators upload and manage the learning materials that students access through the platform. Adaptive Flashcards and Adaptive Test Mode update students' estimated mastery, while AI-assisted learning strategies, AI-generated educational games, and personalized learning activities utilize the current mastery estimates to support independent learning. The Educator Portal provides

a grade-level analytics dashboard that enables teachers to monitor student mastery progress in real time.

The study will be conducted at Pasig Christian Academy involving Junior High School students and their assigned educators, with student respondents selected through Proportionate Stratified Random Sampling. The system will be implemented over a period of 2 to 4 weeks, during which educators will upload learning materials through the Educator Portal while students use the platform as part of their independent learning activities. Data will be collected using researcher-developed questionnaires to evaluate the software quality of LearnSync AI based on the ISO/IEC 25010 Software Quality Model, covering Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

In addition, the correctness of the Bayesian Knowledge Tracing implementation will be verified by comparing the system-generated mastery values with manually computed expected values using predefined test cases. The Generative AI components will also undergo functional verification to ensure that the generated outputs are contextually relevant, factually consistent, and aligned with the uploaded learning resources. Black-box functional testing will likewise be conducted to verify the expected behavior of the platform's major features across both the Educator and Student portals.

Finally, the collected data will be analyzed using appropriate statistical tools, while ethical principles, including compliance with the Data Privacy Act of

2012 (Republic Act No. 10173), will be observed throughout the study. The methodology presented in this chapter provides a structured framework for evaluating the implementation and software quality of LearnSync AI as an independent learning platform that supports the independent learning of Junior High School students under educator-supervised content management.

CHAPTER III

INTRODUCTION

PROJECT CAPABILITIES AND TESTING PROCESS DETAILS

PROJECT TESTING RESULTS

CHAPTER IV

INTRODUCTION

CONCLUSIONS

RECOMMENDATIONS

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APPENDIX A : OPERATIONAL DEFINITION OF TERMS

OPERATIONAL DEFINITION OF TERMS

Adaptive Flashcards - Digital review cards within the platform that adjust reappearance intervals (1, 3, or 7 days) and update BKT mastery probabilities based on a student's self-reported confidence rating (Easy, Medium, or Hard).

Adaptive Test Mode - An automated assessment module that dynamically scales item difficulty (foundational, intermediate, or advanced) based on a student's real-time BKT mastery score and calibrates scoring using format-specific guess probabilities.

Agile Scrum Framework - The iterative software engineering paradigm utilized to structure project management into distinct sprint cycles, backlogs, and rapid testing increments throughout system development.

Bayesian Knowledge Tracing - A probabilistic hidden Markov model implementing four parameters: initial Knowledge, Learning Rate, Slip Probability, and Guess Probability, to continuously track a student's mastery of specific concepts in real time.

Black-Box Functional Testing - A software verification approach used to validate system outputs against specified inputs across the Educator and Student portals without inspecting internal backend source code.

Cognitive Load Theory - An educational design framework that guides the platform's centralized layout to reduce extraneous mental effort caused by tool fatigue and multi-platform navigation.

Cosine Similarity - A mathematical calculation executed natively within the database using the Supabase pgvector extension to measure the semantic

alignment between a student's typed Free Recall summary and the stored lesson embeddings.

Digital Badges - Gamified milestone rewards awarded to students on their progress dashboard upon reaching specific study achievements or mastery thresholds.

Document Mastery Score - The dynamic arithmetic average of all individual concept BKT mastery probabilities within an uploaded document, used to display overall progress and calibrate game difficulty.

Educator Portal - The secure interface enabling authenticated teachers to upload PDF learning modules or textbook scans, assign grade-level viewing access, and monitor student mastery analytics.

Feynman Technique - An AI-assisted study feature that guides an LLM to evaluate student explanations written in plain language by conducting jargon checks, verifying analogy use, and identifying conceptual gaps.

Free Recall - A cognitive study mode where students type out lesson recollections without notes or prompts, evaluated automatically via semantic vector comparison.

Generative Artificial Intelligence - The artificial intelligence pipeline integrated into the platform to generate adaptive test items, flashcards, gamified content, and feedback derived exclusively from educator-uploaded materials.

Guess Probability - A parameter in Bayesian Knowledge Tracing representing the probability that a student answers an item correctly through random

guessing, calibrated dynamically by question format and self-reported confidence level.

Implementation Correctness Rate - A software verification metric that quantifies the percentage of system BKT outputs that match hand-calculated expected values within a +/- 0.01 floating-point tolerance.

Initial Knowledge State - A Bayesian Knowledge Tracing parameter set to a baseline default of 0.10, representing the starting prior probability that a student has mastered a concept before completing any activity.

ISO/IEC 25010 Software Quality Model - The international standard framework used to evaluate LearnSync AI across eight software quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Learning Rate - A Bayesian Knowledge Tracing parameter set to a default value of 0.20, representing the probability that a student transitions from an unmastered to a mastered state following a study opportunity.

LearnSync AI - The web-based personalized learning companion system designed for Junior High School students at Pasig Christian Academy.

Leitner System - A review framework integrated into Adaptive Flashcards that categorizes review items into progressive review tiers based on recall performance.

Optical Character Recognition - The image-processing technology used in the ingestion pipeline to convert scanned printed textbook pages into machine-readable text.

pgvector - An open-source PostgreSQL extension integrated into Supabase to store high-dimensional embeddings and execute vector distance calculations natively within the database tier.

PlayMe Portal - The gamified review hub featuring Crosswords, Term Matching, and Timed Trivia, where game difficulty and timer constraints are calibrated using the student's aggregated document mastery score.

Proportionate Stratified Random Sampling - The probability sampling technique used to select 50 Junior High School student participants proportionally across Grades 7 through 10.

Retrieval Practice - A cognitive study strategy operationalized through quizzes, flashcards, and games that requires learners to reconstruct knowledge from memory rather than passively rereading text.

Row-Level Security - Database policy rules configured in PostgreSQL to isolate individual student data and restrict educator access strictly to their assigned grade-level classes.

Self-Regulated Learning - The process of self-directed study supported by platform analytics, enabling students to set goals, monitor mastery curves, and adjust study habits without direct teacher supervision.

Slip Probability - A Bayesian Knowledge Tracing parameter set to 0.15, representing the probability that a student who has mastered a concept still responds incorrectly due to carelessness, misreading, or fatigue.

Spaced Repetition - A scheduling strategy based on the Leitner system that determines when flashcards reappear for review based on prior recall difficulty.

Student Portal - The web application tier through which Junior High School students access study tools, complete adaptive quizzes, play review games, and view their personalized progress dashboards.

Study Streak - A gamification feature tracking consecutive daily study activity to foster consistent, habit-forming independent learning behavior.

Vector Embeddings - High-dimensional numerical representations of text generated from uploaded study resources, stored in PostgreSQL via pgvector for semantic search and AI retrieval.

APPENDIX B : EVALUATION SHEETS

EVALUATION SHEETS

APPENDIX C : PROJECT TIMELINE

PROJECT TIMELINE

APPENDIX D : WORK ASSIGNMENTS

WORK ASSIGNMENTS

Baruelo, John Louis E. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
1	The Problem and its Background	Crossword	Crossword puzzle based on the concept
1	Significance of the Study	Time Trivia	Quiz based game but with time pressure
1	Statement of the Problem	Matching Card	Matching terms and connect them with each other
2	Software Engineering Paradigm	Admin	Grade Year Level Assignments, Account Creation
2	Population and Sampling		

Leonidas, Ella Mae C. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
1	Review of Related Literature	Student Dashboard	Tracking progress, mastery level, other info
1	Scope and	Educator	Overall tracking

	Limitations	Dashboard	progress, mastery level of each students, other info
2	Evaluation Instrument		
2	Data Collection Method		
2	Ethical Considerations		

Soria, Marc Angelo A. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
2	Introduction	Flashcards	Spaced Repetition System, creation of flashcards
2	Research Design	Feynman	Understand the concept in a simpler one, AI to generate a simpler concept
2	Architectural Framework	Free Recall	Write all of the concepts memorized, AI will check the written concepts
2	Algorithm	Test Mode	Overall understanding of concepts to test

2	Tool to Test for Accuracy	BKT Mastery	Mastery calculations
2	Summary	Educator Upload	Document extraction

APPENDIX E : CERTIFICATE OF APPRECIATION

CERTIFICATE OF APPRECIATION

APPENDIX F : SOURCE CODE

SOURCE CODE

APPENDIX G : PLAGIARISM CHECK RESULTS

PLAGIARISM CHECK RESULTS

APPENDIX H : AI GENERATED CONTENT CHECK RESULTS

AI GENERATED CONTENT CHECK RESULTS

APPENDIX I : EXTENDED ABSTRACT

EXTENDED ABSTRACT

CURRICULUM VITAE